Algebra I: A Self-Contained Course Guide

Through Lecture 13

Built from the course assignments, examples, exercises, and lecture notes

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Lecture roadmap

The chapters follow this sequence of topics through Lecture 13. Some foundational set theory comes first so that the later constructions are self-contained.

Lecture	Topics covered in this book
1	Magmas, semigroups, monoids, groups, identities, inverses, units, powers, order, morphisms.
2	Submagmas, submonoids, subgroups, intersections, generated structures, image/preimage/kernel, products and their universal property.
3	Congruences, quotient magmas/monoids/groups, projection and universal factorization.
4	Free magmas as trees, unique decomposition, recursion and universal property.
5	Free monoids and semigroups as words, free commutative monoids, quotient by commutativity.
6	Free groups, free abelian groups/modules, presentations, group products and free-product coproducts.
7	Quivers, categories, isomorphisms, mono/epi, opposite categories, functors, natural transformations, equivalences, Yoneda, initial/terminal objects, products/coproducts, pullbacks/pushouts.
8	Adjunctions and natural Hom-set bijections.
9	Monoid/group actions, equivariant maps, orbits and their partition.
10	Cosets, index, stabilizers, conjugation, inner automorphisms and conjugacy classes.
11	Orbit–stabilizer, class equation, p-groups and their nontrivial centres.
12	Normal subgroups, quotient groups, kernel characterization, quotient universal property and first isomorphism theorem.
13	Derived subgroup, abelianization, its universal property, functoriality and adjunction.

Chapter 1

Sets and proof language

1.1 Sets, products, and power sets

Definition 1.1. $x \in X$ means that x is an element of X; $A \subseteq X$ means every element of A lies in X. Define

$$A \times B = \{(a, b) : a \in A, b \in B\}, \qquad \mathcal{P}(X) = \{A : A \subseteq X\}.$$

Two sets are equal exactly when they have the same elements.

Intuition

Membership is the basic test for set equality. To prove $A = B$, prove $A \subseteq B$ and $B \subseteq A$. An ordered pair remembers position: $(a, b) = (c, d)$ iff $a = c$ and $b = d$.

Practice 1.2. Prove $(A \cup B)^c = A^c \cap B^c$ inside a fixed set X.

Solution

For $x \in X$, $x \in (A \cup B)^c \iff x \notin A \cup B \iff (x \notin A \text{ and } x \notin B) \iff x \in A^c \cap B^c$.

1.2 Quantifiers and proof patterns

Memorize

$$\neg(\forall x\, P(x)) \iff \exists x\, \neg P(x), \qquad \neg(\exists x\, P(x)) \iff \forall x\, \neg P(x).$$

A counterexample disproves a universal claim. A biconditional needs both directions. To prove $P \Rightarrow Q$, assume P and derive Q; alternatively prove the contrapositive $\neg Q \Rightarrow \neg P$.

Chapter 2

Mappings

2.1 Functions, fibres, and bijections

Definition 2.1. A mapping $f : X \to Y$ assigns each $x \in X$ exactly one $f(x) \in Y$. Its image is $f(X) = \{f(x) : x \in X\}$. The fibre above y is $f^{-1}(\{y\}) = \{x : f(x) = y\}$.

Definition 2.2. $f : X \to Y$ is injective if $f(x) = f(x') \Rightarrow x = x'$; surjective if every $y \in Y$ is some $f(x)$; bijective if both. Equivalently, every fibre has at most one, at least one, or exactly one element, respectively.

Intuition

A function is a machine with declared input and output sets; the formula is not the whole function. Fibres show exactly how inputs are grouped. To prove injection, start with equal outputs. To prove surjection, start with an arbitrary target and construct a preimage.

Practice 2.3. Classify $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$, and then $f : [0, \infty) \to [0, \infty)$.

Solution

On $\mathbb{R}$, $f(1) = f(-1)$ and no negative number is hit, so neither property holds. On $[0, \infty)$, equality of squares implies equality and $t = f(\sqrt{t})$, so it is bijective.

2.2 Composition, inverses, and involutions

If $f : X \to Y$ and $g : Y \to Z$, then $(g \circ f)(x) = g(f(x))$. Composition is associative because both parenthesizations send x to $h(g(f(x)))$. The identity $\mathrm{id}_X(x) = x$ is neutral.

Common trap

Composition generally does not commute. Check sources and targets: one order may not even be defined.

Theorem 2.4. *A map is bijective iff it has a two-sided inverse. That inverse is unique.*

Proof. A bijection has exactly one preimage for each target value; use it to define f^{-1}. Conversely, $g \circ f = \mathrm{id}_X$ gives injection and $f \circ g = \mathrm{id}_Y$ gives surjection. If g, h are inverses, $g = g(fh) = (gf)h = h$. □

A map $s : X \to X$ with $s^2 = \mathrm{id}_X$ is an involution and satisfies $s^{-1} = s$. Complementation $A \mapsto X \setminus A$ on $\mathcal{P}(X)$ is an involution.

Practice 2.5. Let $f : X \to Y$ and $g : Y \to X$. If $g \circ f$ and $f \circ g$ are bijective, prove f, g bijective.

Solution

$g \circ f$ injective implies f injective. $f \circ g$ surjective implies f surjective. Swap f, g for the other result.

2.3 Finite counting and denumerability

If $|A| = n, |B| = m$, there are m^n maps $A \to B$. The number of injections is

$$\frac{m!}{(m-n)!} = m(m-1)\cdots(m-n+1) \quad (n \le m),$$

and 0 when $n > m$. When $n = m$, there are $n!$ bijections.

A set is denumerable if it is in bijection with $\mathbb{N}$.

Practice 2.6. Prove $F : \mathbb{N}^2 \to \mathbb{N}$, $F(a, b) = 2^{a-1}(2b - 1)$, is bijective.

Solution

Every positive integer has a unique expression $2^k q$ with $k \ge 0$ and q positive odd. Put $a = k + 1$ and $q = 2b - 1$. Existence and uniqueness give bijectivity and describe the inverse.

Coursework links

Assignment 1 asks for finite injection counts and this prime-factorization bijection. See also *denumerable sets* and *Factorisation of mappings (how to write proofs)*.

2.4 Permutations that commute

A bijection $X \to X$ is a permutation; all form $\mathrm{Sym}(X)$ under composition.

Proposition 2.7. *For nonempty X, every pair of permutations commutes iff $|X| \le 2$.*

Proof. For one point it is immediate. For two points there are only identity and the swap. If a, b, c are distinct, the swaps (ab) and (bc) do not commute: their two composites send a to different elements. □

Chapter 3

Further tools for sets and mappings

3.1 Images, inverse images, and fibres

Let $f : X \to Y$. For $A \subseteq X$, the *direct image* is

$$f(A) = \{f(a) : a \in A\} \subseteq Y.$$

For $B \subseteq Y$, the *inverse image* is

$$f^{-1}(B) = \{x \in X : f(x) \in B\} \subseteq X.$$

For $y \in Y$, the fibre over y is $f^{-1}(\{y\})$.

Common trap

$f^{-1}(B)$ is meaningful for every function f and every subset $B \subseteq Y$. It does not mean that f has an inverse function. An inverse function exists only when f is bijective.

Direct and inverse images satisfy, for suitable families of subsets,

$$\begin{aligned}
f\left(\bigcup_i A_i\right) &= \bigcup_i f(A_i), \\
f\left(\bigcap_i A_i\right) &\subseteq \bigcap_i f(A_i), \\
f^{-1}\left(\bigcup_i B_i\right) &= \bigcup_i f^{-1}(B_i), \\
f^{-1}\left(\bigcap_i B_i\right) &= \bigcap_i f^{-1}(B_i), \\
f^{-1}(Y \setminus B) &= X \setminus f^{-1}(B).
\end{aligned}$$

The direct-image intersection is only an inclusion because a point in every $f(A_i)$ might be obtained from different preimages in the different sets.

Practice 3.1. Give a function and subsets A, C for which $f(A \cap C)$ is properly contained in $f(A) \cap f(C)$.

Solution

Let $X = \{1, 2\}$, $Y = \{0\}$, and let f be constant. Take $A = \{1\}$ and $C = \{2\}$. Then $A \cap C = \varnothing$, so $f(A \cap C) = \varnothing$, while $f(A) = f(C) = \{0\}$ and hence $f(A) \cap f(C) = \{0\}$.

3.2 The two basic image–preimage inclusions

For every $A \subseteq X$ and $B \subseteq Y$,

$$A \subseteq f^{-1}(f(A)), qquadf(f^{-1}(B)) \subseteq B. \tag{14.1}$$

For the first inclusion, $a \in A$ implies $f(a) \in f(A)$, hence $a \in f^{-1}(f(A))$. For the second, $y \in f(f^{-1}(B))$ means $y = f(x)$ for some x with $f(x) \in B$, so $y \in B$.

The equalities detect injectivity and surjectivity:

$$f^{-1}(f(A)) = A \text{ for every } A \subseteq X \iff f \text{ is injective},$$
$$f(f^{-1}(B)) = B \text{ for every } B \subseteq Y \iff f \text{ is surjective}.$$

For the first converse, apply the equality to $A = \{x\}$: if $f(x') = f(x)$, then $x' \in f^{-1}(f(\{x\})) = \{x\}$. For the second converse, apply it to $B = Y$.

Practice 3.2. Give $f : X \to Y$ and $A \subseteq X$ such that $A \subsetneq f^{-1}(f(A))$.

Solution

Take $X = \{1, 2\}$, $Y = \{0\}$, let f be constant, and take $A = \{1\}$. Then $f(A) = \{0\}$ and $f^{-1}(f(A)) = \{1, 2\}$.

3.3 Left inverses, right inverses, and the shift map

For $f : X \to Y$:

- a *left inverse* is $\ell : Y \to X$ with $\ell f = \mathrm{id}_X$; its existence forces f injective;
- a *right inverse* is $r : Y \to X$ with $fr = \mathrm{id}_Y$; its existence forces f surjective.

Subject to the usual choice of one point in each nonempty fibre, injections admit left inverses when $X \neq \varnothing$, and surjections admit right inverses.

For the shift $s : \mathbb{N} \to \mathbb{N}$, $s(n) = n + 1$, there is no right inverse because 1 is not in its image. It has infinitely many left inverses: for each $k \in \mathbb{N}$, define

$$\ell_k(1) = k, qquad\ell_k(n) = n - 1 \quad (n \geq 2).$$

Then $\ell_k(s(n)) = \ell_k(n + 1) = n$, so $\ell_k s = \mathrm{id}_{\mathbb{N}}$.

3.4 Extending a function from a subset

Let $\varnothing \neq A \subseteq X$ and $f : A \to Y$. Since A is nonempty, Y is nonempty. Choose $y_0 \in Y$ and define

$$g(x) = \begin{cases} f(x), & x \in A, \\ y_0, & x \in X \setminus A. \end{cases}$$

Then $g : X \to Y$ extends f, meaning $g|_A = f$. If the sets are finite, the number of extensions is

$$|Y|^{|X \setminus A|},$$

because each point outside A may be assigned any value of Y independently.

3.5 Equivalence relations and partitions: a counting example

Equivalence relations on a set correspond exactly to partitions: the blocks are the equivalence classes. Therefore the number of equivalence relations on a five-element set is the fifth Bell number.

We can compute it by the number $S(5,k)$ of partitions into k nonempty blocks. The recurrence

$$S(n,k) = S(n-1,k-1) + kS(n-1,k)$$

comes from either putting the new element in its own block or inserting it into one of k existing blocks. For $n = 5$,

$$S(5,1), S(5,2), S(5,3), S(5,4), S(5,5) = 1, 15, 25, 10, 1.$$

Hence the number of equivalence relations is

$$1 + 15 + 25 + 10 + 1 = 52.$$

Chapter 4

The power-set maps f_*, f^*, and $f_!$

4.1 The three definitions

Let $f : X \to Y$. The power sets $\mathcal{P}(X)$ and $\mathcal{P}(Y)$ are ordered by inclusion. The mapping f induces three important maps:

$$\begin{aligned} f_* &: \mathcal{P}(X) \longrightarrow \mathcal{P}(Y), & f_*(A) &= \{f(x) : x \in A\}, \\ f^* &: \mathcal{P}(Y) \longrightarrow \mathcal{P}(X), & f^*(B) &= \{x \in X : f(x) \in B\}, \\ f_! &: \mathcal{P}(X) \longrightarrow \mathcal{P}(Y), & f_!(A) &= \{y \in Y : f^{-1}(\{y\}) \subseteq A\}. \end{aligned}$$

They are called the *lower star*, *upper star*, and *lower shriek*, respectively.

Thus:

- $f_*(A)$ contains the values actually reached from points of A;
- $f^*(B)$ contains the points whose values land in B;
- $f_!(A)$ contains those y for which *every* preimage of y lies in A.

The empty fibre is a subset of every A. Therefore every $y \notin f(X)$ belongs to $f_!(A)$. This often surprises students.

Intuition

Think of a fibre as a vertical column above y. Lower star asks whether at least one point of A appears in the column. Lower shriek asks whether the whole column is contained in A. Upper star collects all columns above a chosen part of Y.

4.2 Complement formula for lower shriek

For $A \subseteq X$,

$$\boxed{f_!(A) = Y \setminus f_*(X \setminus A).} \tag{15.1}$$

Indeed,

$$\begin{aligned} y \in f_!(A) &\iff f^{-1}(\{y\}) \subseteq A \\ &\iff \text{there is no } x \in X \setminus A \text{ with } f(x) = y \\ &\iff y \notin f_*(X \setminus A). \end{aligned}$$

If C_X, C_Y denote complementation, the same identity is

$$f_! = C_Y \circ f_* \circ C_X.$$

4.3 Monotonicity

All three maps preserve inclusion:

$$\begin{aligned} A \subseteq A' &\implies f_*(A) \subseteq f_*(A'), \\ B \subseteq B' &\implies f^*(B) \subseteq f^*(B'), \\ A \subseteq A' &\implies f_!(A) \subseteq f_!(A'). \end{aligned}$$

For the last line, if an entire fibre lies in A, it also lies in the larger set A'.

4.4 What each map preserves

Lower star preserves arbitrary unions:

$$f_*\left(\bigcup_i A_i\right) = \bigcup_i f_*(A_i).$$

It need not preserve intersections unless f is injective.

Upper star preserves arbitrary unions, arbitrary intersections, and complements:

$$\begin{aligned} f^*\left(\bigcup_i B_i\right) &= \bigcup_i f^*(B_i), \\ f^*\left(\bigcap_i B_i\right) &= \bigcap_i f^*(B_i), \\ f^*(Y \setminus B) &= X \setminus f^*(B). \end{aligned}$$

Lower shriek preserves arbitrary intersections:

$$f_!\left(\bigcap_i A_i\right) = \bigcap_i f_!(A_i).$$

To see this, a fibre is contained in every A_i exactly when it is contained in their intersection. Lower shriek need not preserve unions.

4.5 First adjunction: lower star and upper star

For $A \subseteq X$ and $B \subseteq Y$,

$$\boxed{f_*(A) \subseteq B \iff A \subseteq f^*(B).} \tag{15.2}$$

Proof. Assume $f_*(A) \subseteq B$. If $x \in A$, then $f(x) \in f_*(A) \subseteq B$, so $x \in f^*(B)$. Thus $A \subseteq f^*(B)$.

Conversely, assume $A \subseteq f^*(B)$. If $y \in f_*(A)$, then $y = f(x)$ for some $x \in A$. Now $x \in f^*(B)$, so $f(x) = y \in B$. Thus $f_*(A) \subseteq B$. □

This says f_* is the left adjoint of f^* with respect to inclusion.

Useful consequences come from choosing special subsets:

$$A \subseteq f^* f_*(A), qquad f_* f^*(B) \subseteq B.$$

4.6 Second adjunction: upper star and lower shriek

For $A \subseteq X$ and $B \subseteq Y$,

$$\boxed{f^*(B) \subseteq A \iff B \subseteq f_!(A).} \tag{15.3}$$

Proof. Assume $f^*(B) \subseteq A$. Let $y \in B$. If $x \in f^{-1}(\{y\})$, then $f(x) = y \in B$, so $x \in f^*(B) \subseteq A$. Thus the entire fibre over y lies in A, which means $y \in f_!(A)$.

Conversely assume $B \subseteq f_!(A)$. If $x \in f^*(B)$, put $y = f(x)$. Then $y \in B \subseteq f_!(A)$, so $f^{-1}(\{y\}) \subseteq A$. Since x belongs to this fibre, $x \in A$. Therefore $f^*(B) \subseteq A$. □

This says f^* is the left adjoint of $f_!$. Consequences are

$$f^* f_!(A) \subseteq A, qquad B \subseteq f_! f^*(B).$$

4.7 Composition and identities

For $X \xrightarrow{f} Y \xrightarrow{g} Z$,

$$\begin{aligned}(g \circ f)_* &= g_* \circ f_*,\\ (g \circ f)^* &= f^* \circ g^*,\\ (g \circ f)_! &= g_! \circ f_!.\end{aligned}$$

Notice the reversal in the upper-star formula: inverse images are contravariant. For identity mappings, all three induced maps are identity maps on the relevant power set.

4.8 Detecting injections and surjections with f^*

Let $F = f^* : \mathcal{P}(Y) \to \mathcal{P}(X)$.

Theorem 4.1. *f is injective iff F is surjective; f is surjective iff F is injective. Consequently, f is bijective iff F is bijective.*

Proof. If f is injective and $A \subseteq X$, then $A = f^{-1}(f(A)) = F(f_*(A))$, so every subset of X lies in the image of F.

Conversely, suppose F is surjective. For each $A \subseteq X$, choose B with $A = f^{-1}(B)$. Inverse images are unions of complete fibres. Taking $A = \{x\}$ shows the fibre through x contains only x, so f is injective.

If f is surjective and $F(B) = F(C)$, applying f_* gives $B = f_* f^*(B) = f_* f^*(C) = C$, so F is injective.

Conversely, if F is injective, compare Y and $f(X)$. Both have inverse image X: $f^*(Y) = X = f^*(f(X))$. Hence $Y = f(X)$, so f is surjective. □

4.9 Worked calculation: the square map

Let $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$. Then

$$\begin{aligned}f^*([-2,8)) &= \{x : -2 \le x^2 < 8\} = (-2\sqrt{2}, 2\sqrt{2}),\\ f^*((-\infty, 0]) &= \{0\},\\ f^*((-1,1)) &= (-1,1),\\ f^*(\mathbb{Z}) &= \{x \in \mathbb{R} : x^2 \in \mathbb{Z}\} = \{\pm\sqrt{n} : n \in \mathbb{Z}_{\ge 0}\}.\end{aligned}$$

For the first line, $x^2 \ge 0$ automatically, while $x^2 < 8$ is equivalent to $|x| < 2\sqrt{2}$.

4.10 A finite example containing all three maps

Let $X = \{1, 2, 3\}$, $Y = \{a, b, c\}$, with

$$f(1) = a, \qquad f(2) = a, \qquad f(3) = b.$$

Take $A = \{1, 3\}$ and $B = \{a, c\}$. Then

$$\begin{aligned} f_*(A) &= \{a, b\}, \\ f^*(B) &= \{1, 2\}, \\ f_!(A) &= \{b, c\}. \end{aligned}$$

The fibre over a is $\{1, 2\}$, not contained in A; the fibre over b is $\{3\} \subseteq A$; the fibre over c is empty and hence contained in A.

Checkpoint: Power-set maps

Before moving on, be able to state all three definitions from memory, prove both adjunctions element-by-element, derive the complement formula for $f_!$, calculate all three maps in a finite example, and prove the injection/surjection theorem for f^*.

Chapter 5

Relations and quotients

5.1 Equivalence relations

A relation on X is a subset of $X \times X$. It is an equivalence relation if it is reflexive ($x \sim x$), symmetric ($x \sim y \Rightarrow y \sim x$), and transitive ($x \sim y \sim z \Rightarrow x \sim z$). The class of x is $[x] = \{y : y \sim x\}$ and $X/\sim = \{[x] : x \in X\}$.

Intuition

An equivalence relation consistently declares objects the same for the present purpose. Its classes are disjoint buckets partitioning X. Conversely, every partition defines an equivalence relation.

Theorem 5.1. $[x] = [y]$ *iff* $x \sim y$. *Hence two classes are equal or disjoint.*

Proof. If $x \sim y$ and $z \sim x$, transitivity gives $z \sim y$; symmetry gives the reverse inclusion. Conversely $x \in [x] = [y]$ gives $x \sim y$. □

5.2 Classes as fibres of an invariant

If $I : X \to S$, then $x \sim y \iff I(x) = I(y)$ is automatically an equivalence relation, and its classes are fibres of I.

Practice 5.2. On $\mathbb{R}$, let $x \sim y$ iff $x - y \in \mathbb{Q}$. Prove equivalence and describe $[x]$.

Solution

Use closure of $\mathbb{Q}$ under 0, negatives, and sums. The class is $[x] = x + \mathbb{Q} = \{x + q : q \in \mathbb{Q}\}$.

Practice 5.3. On $X = \mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$, let $(a, b) \sim (c, d)$ iff $ad = bc$. Prove equivalence.

Solution

Use the invariant $I(a, b) = a/b \in \mathbb{Q}$. Cross multiplication says exactly $I(a, b) = I(c, d)$, so the relation is equivalence. Its classes are all fraction representations of one rational.

Practice 5.4. On $\mathbb{N}^2$, let $(a, b) \sim (c, d)$ iff $a + d = b + c$. Describe the classes.

Solution

Rearrange to $a - b = c - d$. The invariant is $a - b \in \mathbb{Z}$, and $C_k = \{(a, b) : a - b = k\}$.

Other Assignment 1 patterns include congruence $a \sim b \iff n \mid (a-b)$, with residue classes, and scaling by powers of 2 on nonzero integers, with orbit $\{2^k a : k \in \mathbb{Z}\}$ restricted to integral values.

5.3 Quotient factorisation

The projection $\pi : X \to X/\!\sim$, $x \mapsto [x]$, is surjective.

Theorem 5.5 (universal property)**.** *If $f : X \to Y$ is constant on classes, there is a unique $\bar{f} : X/\!\sim \to Y$ with $f = \bar{f} \circ \pi$. It is $\bar{f}([x]) = f(x)$.*

Proof. The hypothesis makes the formula independent of the representative. It gives the desired equality, and surjectivity of π forces uniqueness. □

Common trap

Any definition using representatives must first be proved well-defined. Replace every representative by an equivalent one and verify the output is unchanged.

Chapter 6

Binary operations

Definition 6.1. A law of composition on S is a map $* : S \times S \to S$. The pair $(S, *)$ is a magma. Closure is part of this definition. The operation is associative if $(a * b) * c = a * (b * c)$ and commutative if $a * b = b * a$. An associative magma is a semigroup.

Intuition

A binary operation is a multiplication table, possibly infinite. To test associativity, compute both parenthesizations with independent variables. To disprove it, one triple suffices.

Practice 6.2. On $\mathbb{R}$, let $a * b = a + ab$. Test associativity and commutativity.

Solution

$a * b = a + ab$ while $b * a = b + ab$, so not commutative. Also $(a * b) * c = a + ab + ac + abc$ while $a * (b * c) = a + ab + abc$, so not associative.

Practice 6.3. On X, let $a * b = a$. Test both properties.

Solution

Both parenthesizations equal a, so associative. It is commutative exactly when $|X| \leq 1$.

6.1 Counting operations

If $|S| = n$, there are n^{n^2} operations because an operation is a function $S^2 \to S$. A commutative table is determined by its $n(n+1)/2$ diagonal-and-upper-triangle cells, giving $n^{n(n+1)/2}$ commutative operations.

Practice 6.4. Disprove: every commutative operation on a two-element set is associative.

Solution

On $\{0, 1\}$ set $0 * 0 = 1$, $0 * 1 = 1 * 0 = 0$, $1 * 1 = 0$. It is commutative, but $(0 * 0) * 1 = 0$ while $0 * (0 * 1) = 1$.

Chapter 7

Monoids and groups

Definition 7.1. An identity e satisfies $e * a = a = a * e$. A monoid is an associative magma with identity. A group is a monoid in which each a has an inverse a^{-1} with $aa^{-1} = e = a^{-1}a$. It is abelian if commutative.

Identity and inverses are unique. Cancellation holds. Also $(ab)^{-1} = b^{-1}a^{-1}$ and $ax = b$, $ya = b$ have unique solutions $a^{-1}b$, ba^{-1}.

Intuition

A group is a world of reversible actions: identity does nothing, inverses undo, and associativity lets a chain of actions be composed unambiguously.

Common trap

In nonabelian groups order matters: $(ab)^{-1} = b^{-1}a^{-1}$. Never commute factors without permission.

Any semigroup S can acquire a new formal identity $e \notin S$: retain the old product on S and declare $e + a = a = a + e$. Associativity follows because if an entry is e it may be deleted; if all entries lie in S, use old associativity. This is Assignment 2, Problem 4.

Practice 7.2. Is $(\mathbb{Z}, *)$, $a * b = a + b + ab$, a group?

Solution

It is associative because $1 + (a * b) = (1 + a)(1 + b)$, and identity is 0. But $1 * x = 0$ requires $x = -1/2 \notin \mathbb{Z}$, so it is not a group.

Practice 7.3. Is $a * b = a/b$ on $\mathbb{R}_{>0}$ a group?

Solution

No: $(a * b) * c = a/(bc)$ but $a * (b * c) = ac/b$ generally.

7.1 Roots of unity and pointwise groups

For $\mu_n = \{z \in \mathbb{C} : z^n = 1\}$, closure follows from $(zw)^n = z^n w^n = 1$, identity is 1, and $z^{-1} = z^{n-1}$. Associativity is inherited from $\mathbb{C}$, so μ_n is an abelian group.

For a group G and nonempty set X, the functions G^X form a group pointwise:

$$(fg)(x) = f(x)g(x), \quad \mathbf{e}(x) = e_G, \quad f^{-1}(x) = f(x)^{-1}.$$

Check every axiom at an arbitrary x.

Practice 7.4. Prove pointwise associativity.

Solution

$((fg)h)(x) = (f(x)g(x))h(x) = f(x)(g(x)h(x)) = (f(gh))(x)$ for every x.

Coursework links

This is Assignment 2, Problem 7. The sheet *product_magma* develops the same componentwise method.

Chapter 8

Substructures, morphisms, and congruences

We now collect the structural results developed in Lectures 1–3. The definitions and examples above give the concrete starting point for these general constructions.

8.1 The hierarchy and its maps

Definition 8.1. A *magma* is a set M with a binary operation $M \times M \to M$. A *semigroup* is an associative magma. A *monoid* is a semigroup with an identity e. A *group* is a monoid in which every element is invertible. A *commutative monoid* is a monoid with $xy = yx$. For magmas M, N, a *magma morphism* $f : M \to N$ satisfies $f(xy) = f(x)f(y)$. A *monoid morphism* also satisfies $f(e_M) = e_N$. A group morphism is a magma morphism between groups; it automatically preserves identities and inverses. An *isomorphism* is a morphism with a two-sided inverse morphism.

Intuition

A morphism preserves the structure we have named. A monoid has a distinguished identity, so its morphisms must preserve that point as well as multiplication. In a group the identity condition follows from inverses, but in a general monoid it does not.

Practice 8.2. Give a magma morphism between monoids that is not a monoid morphism. Prove that every group morphism preserves identity and inverses.

Solution

The constant map $f : (\mathbb{Z}, +) \to (\mathbb{Z}, \cdot)$, $f(n) = 0$, preserves the binary laws because $0 = 0 \cdot 0$, but sends the source identity 0 to 0, not the target identity 1. For a group morphism h, $h(e) = h(e)^2$; multiplying by $h(e)^{-1}$ gives $h(e) = e$. Then $e = h(e) = h(xx^{-1}) = h(x)h(x^{-1})$, so $h(x^{-1}) = h(x)^{-1}$.

Composition of morphisms is a morphism. The inverse of a bijective magma, monoid, or group morphism preserves the corresponding operation: if $u = f(x), v = f(y)$, then $f^{-1}(uv) = f^{-1}(f(xy)) = xy$. Identities are also preserved where required. Thus "bijective morphism" and "isomorphism" agree here.

8.2 Identity, units, powers, and generated substructures

Definition 8.3. An element u of a monoid M is a *unit* if some $v \in M$ satisfies $uv = vu = e$. Its inverse v is unique. The set $M^\times$ of units is a group. For $x \in M$, define $x^0 = e$ and $x^{n+1} = x^n x$.

If x is a unit, put $x^{-n} = (x^{-1})^n$. The *order* of g in a group is the least positive n with $g^n = e$, if such n exists, and is infinite otherwise.

Associativity gives $x^{m+n} = x^m x^n$ for nonnegative integers, and for all integers when x is a unit. The map $\mathbb{Z} \to G$, $n \mapsto g^n$, is a group morphism; its image is the cyclic subgroup $\langle g \rangle$. The group $M^\times$ is closed since $(uv)^{-1} = v^{-1}u^{-1}$. For any monoid morphism f, $f(M^\times) \subseteq N^\times$.

Practice 8.4. Find the units of $(\mathbb{Z}, \cdot)$ and the order of $[3]$ in $(\mathbb{Z}/12\mathbb{Z}, +)$.

Solution

An integer u has an integer multiplicative inverse only if $u = 1$ or $u = -1$, so $\mathbb{Z}^\times = \{1, -1\}$. Additively, $n[3] = [0]$ iff $12 \mid 3n$, equivalently $4 \mid n$, so the order is 4.

8.3 Submagmas, submonoids, and subgroups

Definition 8.5. For a magma M, a subset $S \subseteq M$ is a *submagma* if $x, y \in S \Rightarrow xy \in S$; the empty subset is allowed if one permits empty magmas. For a monoid M, a *submonoid* S must be closed under multiplication *and contain the ambient identity* e_M. For a group G, a *subgroup* H is a submonoid containing h^{-1} whenever $h \in H$.

Intuition

Each test adds one demand: closure, then the inherited identity, then inherited inverses. The word "inherited" matters; a subset's internal identity need not be the identity of the ambient monoid.

Practice 8.6. Decide which of $2\mathbb{Z} \subseteq (\mathbb{Z}, +)$, $\mathbb{N} \subseteq (\mathbb{Z}, +)$, and $\{0, 1\} \subseteq (\mathbb{Z}, \cdot)$ are submonoids and subgroups. Recall $\mathbb{N} = \{1, 2, \dots\}$.

Solution

$2\mathbb{Z}$ contains 0, is closed under addition, and contains additive inverses, so is a subgroup. $\mathbb{N}$ is closed under addition but misses the ambient identity 0, so is only a submagma/semigroup. $\{0, 1\}$ contains multiplicative identity 1 and is closed under multiplication, so is a submonoid; 0 has no inverse, so it is not a subgroup.

An arbitrary intersection of submagmas is a submagma; an arbitrary intersection of submonoids is a submonoid; and an arbitrary intersection of subgroups is a subgroup (including an empty family, whose intersection is the whole structure). This works because each defining closure condition survives intersection.

Definition 8.7. The substructure $\langle T \rangle$ *generated* by $T \subseteq M$ is the intersection of all substructures containing T. It is the smallest such substructure. In a monoid it consists of all finite products of elements of T, including the empty product e. In a group it consists of finite products of elements of $T \cup T^{-1}$, including e.

For a single element, a monoid gives $\langle x \rangle_{\text{mon}} = \{e, x, x^2, \dots\}$ while a group gives $\langle x \rangle_{\text{grp}} = \{x^n : n \in \mathbb{Z}\}$. The former need not be a group. More generally a generated *magma* consists of all parenthesized finite expressions in T; without associativity, parentheses cannot be discarded.

Practice 8.8. Compute the submonoid generated by 2 in $(\mathbb{Z}, +)$ and the subgroup generated by 2 in the same ambient group.

Solution

Finite sums of 2, including the empty sum, give $\{0, 2, 4, 6, \dots\}$. Allowing inverses gives $\{2n : n \in \mathbb{Z}\} = 2\mathbb{Z}$.

8.4 Images, inverse images, and kernels

If $f : M \to N$ is a morphism, the image of a submagma is a submagma; for monoid morphisms the image of a submonoid is a submonoid; for group morphisms the image of a subgroup is a subgroup. The inverse image of a substructure of N is a corresponding substructure of M. For example, if $K \leq N$ and $a, b \in f^{-1}(K)$, then $f(ab^{-1}) = f(a)f(b)^{-1} \in K$, so $ab^{-1} \in f^{-1}(K)$. The image of a generated substructure satisfies $f(\langle T \rangle) = \langle f \rangle (T)\rangle$ with the relevant type of generation.

Definition 8.9. For a monoid morphism $f : M \to N$, $\ker f = f^{-1}(\{e_N\})$ is a submonoid. For a group morphism it is a subgroup. A magma alone has no distinguished identity, so a kernel in this sense is not defined.

Common trap

For a general monoid morphism, a trivial kernel does *not* imply injectivity. The group proof uses inverses, which monoids may lack.

Practice 8.10. Show that $f : (\mathbb{N}_0, +) \to (\{0, 1\}, \max)$ defined by $f(0) = 0$ and $f(n) = 1$ for $n > 0$ is a noninjective monoid morphism with trivial kernel.

Solution

Here $\mathbb{N}_0 = \{0, 1, 2, \dots\}$ and 0 is the identity for both monoids. The sum $m + n$ is zero exactly when both summands are zero, so $f(m + n) = \max(f(m), f(n))$. Its kernel is $\{0\}$, but $f(1) = f(2)$.

8.5 Products, projections, and uniqueness

For magmas M, N, define $(m, n)(m', n') = (mm', nn')$. Associativity, identities, inverses, and commutativity hold componentwise when they hold in both factors. The projections $p_1 : M \times N \to M$, $p_2 : M \times N \to N$ are morphisms. For morphisms $f : P \to M$, $g : P \to N$, the unique morphism with $p_1 h = f$ and $p_2 h = g$ is $h(p) = (f(p), g(p))$. Uniqueness follows because a pair is fixed by its two coordinates. Any two objects satisfying this same product property are uniquely isomorphic by sending each through the other's universal arrows.

Practice 8.11. Prove that the group inverse of $(g, h) \in G \times H$ is (g^{-1}, h^{-1}).

Solution

Componentwise multiplication gives $(g, h)(g^{-1}, h^{-1}) = (gg^{-1}, hh^{-1}) = (e_G, e_H)$ and the product in the opposite order is the same identity.

8.6 Congruences and quotient structures

Definition 8.12. A *congruence* on a magma M is an equivalence relation $\sim$ compatible with multiplication: $x \sim x'$ and $y \sim y'$ imply $xy \sim x'y'$. Then $M/\sim$ has the well-defined operation $[x][y] = [xy]$.

Intuition

An equivalence relation only groups elements; a congruence also permits multiplication of groups of elements. This is exactly the representative-independence check.

The quotient projection $\pi : M \to M/\sim$ is a surjective magma morphism. Associativity and commutativity descend; if M is a monoid, $[e]$ is the quotient identity; if M is a group, $[x^{-1}]$ is the inverse of $[x]$. The quotient operation is the unique one making π a morphism, since surjectivity forces $[x][y] = \pi(xy)$. For any morphism $f : M \to N$, the relation $x \sim_f y \iff f(x) = f(y)$ is a congruence. Conversely, a map $h : M \to P$ factors uniquely as $h = \bar{h}\pi$ through $M/\sim$ exactly when $x \sim y \Rightarrow h(x) = h(y)$; if h is a morphism, so is $\bar{h}$.

Practice 8.13. For $n \geq 1$, show that congruence modulo n is a congruence on $(\mathbb{Z}, +)$ and identify the quotient operation.

Solution

If $a - a' = nr$ and $b - b' = ns$, then $(a + b) - (a' + b') = n(r + s)$, so $a + b \equiv a' + b' \pmod{n}$. Thus $[a] + [b] = [a + b]$ is well-defined. Identity is $[0]$ and inverse of $[a]$ is $[-a]$.

Checkpoint: Lectures 1–3

Given a subset, distinguish submagma, submonoid, and subgroup; compute its generated structure. Given a map, check operation and identity separately. Given a quotient, prove congruence before defining the quotient law.

Chapter 9

Concrete groups, subgroups, and cyclic groups

9.1 Affine transformations

For $a \neq 0$, set $f_{a,b}(x) = ax + b$. Then

$$f_{a,b} \circ f_{c,d} = f_{ac,ad+b}, \quad \mathrm{id} = f_{1,0}, \quad f_{a,b}^{-1} = f_{1/a,-b/a}.$$

Thus these maps form a group under composition. It is generally nonabelian.

9.2 The subgroup test

Theorem 9.1. *A nonempty $H \subseteq G$ is a subgroup iff $a, b \in H \Rightarrow ab^{-1} \in H$. Additively, require $a - b \in H$.*

Proof. Necessity is clear. Conversely choose $h \in H$: $e = hh^{-1} \in H$. Taking $a = e$ gives inverses, then $ab = a(b^{-1})^{-1}$ gives closure. Associativity is inherited. □

Practice 9.2. Classify Assignment 3, Problem 4.

Solution

(i) $\mathbb{R}_{>0} \not\leq (\mathbb{R}, +)$: no 0 or negatives. (ii) $\mathbb{R}_{>0} \leq (\mathbb{R}^\times, \cdot)$. (iii) $\mathbb{Q}_{>0} \leq (\mathbb{R}_{>0}, \cdot)$. (iv) $\pi\mathbb{Q} \leq (\mathbb{R}, +)$ since $\pi q - \pi r = \pi(q - r)$.

For $x \in X$, $A_x = \{f \in \mathrm{Sym}(X) : f(x) = x\}$ is a subgroup: identity, composites, and inverses fix x. If $x \neq y$, $A_{x,y} = \{f : f(x) = y\}$ is not a subgroup because it misses the identity.

9.3 Generated cyclic subgroups

$\langle g \rangle = \{g^k : k \in \mathbb{Z}\}$; additively, $\langle a \rangle = \{ka : k \in \mathbb{Z}\}$. Thus

$$\begin{aligned}
\langle i \rangle &= \{1, i, -1, -i\}, \\
\left\langle \sqrt{2} \right\rangle \text{ in } (\mathbb{R}_{>0}, \cdot) &= \{2^{k/2} : k \in \mathbb{Z}\}, \\
\left\langle \sqrt{2} \right\rangle \text{ in } (\mathbb{R}, +) &= \{k\sqrt{2} : k \in \mathbb{Z}\}, \\
\left\langle -\sqrt{2} \right\rangle \text{ in } (\mathbb{C}^\times, \cdot) &= \{(-\sqrt{2})^k : k \in \mathbb{Z}\}, \\
\langle 12 \rangle \text{ in } (3\mathbb{Z}, +) &= 12\mathbb{Z}.
\end{aligned}$$

The ambient operation decides between powers and multiples.

Chapter 10

Homomorphisms and isomorphisms

Definition 10.1. $\phi : G \to H$ is a homomorphism if $\phi(ab) = \phi(a)\phi(b)$. It is an isomorphism if also bijective. Define $\ker \phi = \{g : \phi(g) = e\}$ and $\operatorname{im} \phi = \{\phi(g) : g \in G\}$.

Every homomorphism preserves identity, inverses, and integer powers. Kernel and image are subgroups; ϕ is injective iff $\ker \phi = \{e\}$.

Practice 10.2. Classify Assignment 3, Problem 7.

Solution

(i) $a \mapsto 2a$ on $\mathbb{Z}$ is a homomorphism. (ii) cos is not: it sends 0 to 1. (iii) $A \mapsto A \cap Y$ preserves symmetric difference (check membership). (iv) relative complement does not: it sends $\varnothing$ to Y. (v) evaluation $\alpha \mapsto \alpha(x_0)$ preserves pointwise addition.

10.1 Maps from $\mathbb{Z}$ and the cube map

If $f : (\mathbb{Z}, +) \to (\mathbb{R}, +)$ and $a = f(1)$, then $f(n) = na$ for positive, zero, and negative n. Conversely $f_a(n) = an$ is a homomorphism. This solves Problem 8.

Suppose $f(x) = x^3$ is an injective homomorphism. First compare the two expressions for $(yx)^3$. Since $yx = x^{-1}(xy)x$,

$$(yx)^3 = x^{-1}(xy)^3 x = x^2 y^3 x,$$

while the homomorphism property gives $(yx)^3 = y^3 x^3$. Cancelling the rightmost x shows $x^2 y^3 = y^3 x^2$. Thus every square commutes with every cube.

In particular x^6 commutes with y^6. But $x^6 = f(x^2)$ and $y^6 = f(y^2)$, so

$$f(x^2 y^2) = f(y^2 x^2).$$

Injectivity gives $x^2 y^2 = y^2 x^2$. Hence x^2 commutes with both y^2 and y^3, and therefore with $y = y^3 (y^2)^{-1}$. Every square is central. Finally,

$$(xy)^3 = x^3 y^3 \quad \Longrightarrow \quad (xy)^2 xy = x^2 y^2 xy,$$

where centrality of squares was used on the right. Cancel xy and then cancel the outer x, y from $xyxy = xxyy$ to obtain $yx = xy$. Thus G is abelian. This proof also shows exactly where injectivity is needed.

10.2 Exponential isomorphism

$\exp : (\mathbb{R}, +) \to (\mathbb{R}_{>0}, \cdot)$ is a bijective homomorphism with inverse log. But $(\mathbb{Q}, +)$ and $(\mathbb{Q}_{>0}, \cdot)$ are not isomorphic: the first is divisible (every q has an additive nth root q/n), while 2 has no rational multiplicative square root in the second.

Chapter 11

Further group theory and practice

11.1 Fast hierarchy and terminology

Keep the hierarchy visible:

$$\text{magma} \ + \ \text{associativity} = \text{semigroup},$$

$$\text{semigroup} \ + \ \text{identity} = \text{monoid},$$

$$\text{monoid} \ + \ \text{inverse for every element} = \text{group}.$$

In the corrected practice-sheet terminology, a unit is an invertible element of a monoid. In the modified-product problem, however, "a unit relative to the new product" is explicitly asking for a neutral element for that new associative law.

11.2 Rotations about the origin

Every rotation about the origin has an angle θ and may be denoted R_θ. Composition satisfies

$$R_\theta R_\phi = R_{\theta+\phi}.$$

Closure follows because the composite is another rotation about the origin. Composition is associative. The identity is R_0, and $R_\theta^{-1} = R_{-\theta}$. Hence all plane rotations about the origin form an abelian group.

The map

$$\mathbb{R}/2\pi\mathbb{Z} \longrightarrow G, \qquad [\theta] \longmapsto R_\theta$$

is a well-defined isomorphism: angles differing by an integer multiple of 2π describe the same rotation.

11.3 Cyclic groups and subgroups of $\mathbb{Z}_n$

Write $\mathbb{Z}_n = (\mathbb{Z}/n\mathbb{Z}, +_n)$. For $[d] \in \mathbb{Z}_n$,

$$\langle [d] \rangle = \{[kd] : k \in \mathbb{Z}\}.$$

Its order is

$$\operatorname{ord}([d]) = \frac{n}{\gcd(n,d)}. \tag{16.1}$$

Indeed, $k[d] = [0]$ exactly when $n \mid kd$, and the least positive such k is $n/\gcd(n,d)$.

Every subgroup of a cyclic group is cyclic. The subgroups of $\mathbb{Z}_n$ correspond to divisors of n: for each $d \mid n$, the subgroup $\langle [d] \rangle$ has order n/d.

For example, the subgroups of $\mathbb{Z}_{24}$ are

$$\langle [1] \rangle, \langle [2] \rangle, \langle [3] \rangle, \langle [4] \rangle, \langle [6] \rangle, \langle [8] \rangle, \langle [12] \rangle, \langle [0] \rangle.$$

Their orders are $24, 12, 8, 6, 4, 3, 2, 1$. Since 7 is prime, $\mathbb{Z}_7$ has only $\{[0]\}$ and $\mathbb{Z}_7$.

Practice-sheet computations follow immediately:

$$\begin{aligned}
&\langle [7] \rangle \leq \mathbb{Z}_{18} = \mathbb{Z}_{18} && (\gcd(7,18) = 1),\\
&\langle [5] \rangle \leq \mathbb{Z}_{20} = \{[0],[5],[10],[15]\},\\
&\langle [3] \rangle \leq \mathbb{Z}_{12} = \{[0],[3],[6],[9]\},\\
&\langle [3] \rangle \leq \mathbb{Z}_{16} = \mathbb{Z}_{16} && (\gcd(3,16) = 1).
\end{aligned}$$

11.4 Conjugate subgroups

If $H \leq G$ and $a \in G$, then

$$aHa^{-1} = \{aha^{-1} : h \in H\}$$

is a subgroup. It is nonempty because $aea^{-1} = e$. If aha^{-1}, aka^{-1} belong to it, then

$$(aha^{-1})(aka^{-1})^{-1} = aha^{-1}ak^{-1}a^{-1} = a(hk^{-1})a^{-1} \in aHa^{-1}.$$

The subgroup test completes the proof. The map $h \mapsto aha^{-1}$ is actually an isomorphism $H \to aHa^{-1}$.

11.5 Homomorphisms from cyclic groups

A homomorphism from a cyclic group is determined by the image of one generator.

For $\phi : \mathbb{Z}_m \to \mathbb{Z}_n$, put $\phi([1]_m) = [a]_n$. The definition is possible exactly when

$$m[a]_n = [0]_n, \quad \text{equivalently} \quad n \mid ma. \tag{16.2}$$

Then $\phi([k]_m) = [ka]_n$. It is injective exactly when $[a]_n$ has order m.

Examples:

- An injection $\mathbb{Z}_8 \to \mathbb{Z}_{24}$ is $[k]_8 \mapsto [3k]_{24}$, because $[3]_{24}$ has order 8.
- A homomorphism $\mathbb{Z}_7 \to \mathbb{Z}_{15}$ must send $[1]$ to $[a]$ with $7a \equiv 0 \pmod{15}$. Since 7 is invertible modulo 15, $a \equiv 0$; there is exactly one homomorphism.
- If $a \in G$, the formula $[i]_n \mapsto a^i$ is a homomorphism precisely when $a^n = e$. Well-definedness requires $a^{i+n} = a^i$, which is equivalent to $a^n = e$.

The roots of unity satisfy

$$\mathbb{Z}_n \cong \mu_n, qquad[k] \longmapsto e^{2\pi ik/n}.$$

The map is well-defined, multiplicative with respect to addition in $\mathbb{Z}_n$, and bijective because these are exactly the n distinct roots.

11.6 Quick homomorphism tests

Before calculating, check that the proposed formula is well-defined on equivalence classes and sends the identity to the identity.

For the practice-sheet examples:

- $\mathbb{Z} \to \mathbb{Z}$, $a \mapsto 2a$, is a homomorphism.

- The rule $\mathbb{Z}_6 \to \mathbb{Z}_8$, $[a]_6 \mapsto [a]_8$, is not well-defined because $[0]_6 = [6]_6$ but $[0]_8 \neq [6]_8$.
- The rule $\mathbb{Z}_6 \to \mathbb{Z}$, $[a]_6 \mapsto a$, is likewise not well-defined.
- Parity $\mathbb{Z}_8 \to \mathbb{Z}_2$ is a well-defined surjective homomorphism.
- Parity $\mathbb{Z}_{15} \to \mathbb{Z}_2$ is not well-defined because $[0]_{15} = [15]_{15}$ but 0 and 15 have different parity.

For $\mathbb{C}^\times \to \mathbb{C}^\times$:

- $z \mapsto \bar{z}$ is a homomorphism because $\overline{zw} = \bar{z}\bar{w}$;
- $z \mapsto z^2$ is a homomorphism because $\mathbb{C}^\times$ is abelian;
- $z \mapsto iz$ is not, since it sends 1 to $i \neq 1$;
- $z \mapsto |z|$ is a homomorphism, with its positive real value regarded as a nonzero complex number.

The sign map $\mathbb{R}^\times \to \{1, -1\}$ is a surjective homomorphism because $\operatorname{sgn}(ab) = \operatorname{sgn}(a)\operatorname{sgn}(b)$ and both signs occur.

11.7 Two useful homomorphism criteria

Proposition 11.1. *A mapping $f : G \to H$ is a group homomorphism iff*

$$f(ab^{-1}) = f(a)f(b)^{-1}$$

for every $a, b \in G$.

Proof. Necessity follows from preservation of products and inverses. Conversely, put $a = b$ to get $f(e) = e$. Put $a = e$ to obtain $f(b^{-1}) = f(b)^{-1}$. Finally replace b by b^{-1}:

$$f(ab) = f(a)f(b^{-1})^{-1} = f(a)f(b).$$

□

Proposition 11.2. *$\phi : G \to H$ is a homomorphism iff its graph*

$$\Gamma_\phi = \{(g, \phi(g)) : g \in G\}$$

is a subgroup of $G \times H$.

Proof. If ϕ is a homomorphism, the subgroup test gives

$$(g, \phi(g))(k, \phi(k))^{-1} = (gk^{-1}, \phi(g)\phi(k)^{-1}) = (gk^{-1}, \phi(gk^{-1})) \in \Gamma_\phi.$$

Conversely, if the graph is a subgroup, the product $(g, \phi(g))(k, \phi(k)) = (gk, \phi(g)\phi(k))$ belongs to the graph. Its first coordinate forces its second coordinate to equal $\phi(gk)$, so $\phi(gk) = \phi(g)\phi(k)$. □

11.8 No surjection $\mathbb{Z} \to \mathbb{R}$ and automorphisms of $\mathbb{Z}$

Every homomorphism $f : (\mathbb{Z}, +) \to (\mathbb{R}, +)$ is determined by $a = f(1)$ and satisfies $f(n) = na$. Its image $a\mathbb{Z}$ is countable, whereas $\mathbb{R}$ is uncountable, so it cannot be surjective.

For an isomorphism $f : \mathbb{Z} \to \mathbb{Z}$, surjectivity requires $f(1) = a$ to generate all of $\mathbb{Z}$, so $a = 1$ or $a = -1$. Therefore the only automorphisms are

$$n \mapsto n, qquadn \mapsto -n.$$

11.9 Power maps and commutativity

If every $g \in G$ satisfies $g^2 = e$, then $g^{-1} = g$. For $a, b \in G$,

$$ab = (ab)^{-1} = b^{-1}a^{-1} = ba,$$

so G is abelian.

The inversion map $\iota(g) = g^{-1}$ is a homomorphism iff G is abelian. Indeed,

$$\iota(ab) = b^{-1}a^{-1}, qquad \iota(a)\iota(b) = a^{-1}b^{-1},$$

and equality for all a, b is equivalent to commutativity.

Likewise, $q(g) = g^2$ is a homomorphism iff G is abelian. If it is a homomorphism,

$$abab = (ab)^2 = a^2b^2 = aabb,$$

and cancellation gives $ba = ab$. The converse follows by commuting a, b.

If G is abelian, the set $G^n = \{g^n : g \in G\}$ is a subgroup because

$$g^n(h^n)^{-1} = g^n h^{-n} = (gh^{-1})^n.$$

It can fail in a nonabelian group. For $n = 3$ in S_3, the cubes are the identity and the three transpositions. The product of two distinct transpositions is a 3-cycle, which is not a cube in S_3, so the set of cubes is not a subgroup.

The condition $g^3 = e$ for every g does not force commutativity. A concrete counterexample is the group of upper-unitriangular matrices over $\mathbb{F}_3$:

$$\left\{ \begin{pmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix} : a, b, c \in \mathbb{F}_3 \right\}.$$

Every element has cube equal to the identity, but matrices with $(a, b) = (1, 0)$ and $(0, 1)$ do not commute.

11.10 Product groups and common isomorphism invariants

For groups G, H, the product group has

$$(g, h)(g', h') = (gg', hh'), \quad e = (e_G, e_H), \quad (g, h)^{-1} = (g^{-1}, h^{-1}).$$

The map

$$\mathbb{C} \longrightarrow \mathbb{R} \times \mathbb{R}, qquad a + bi \longmapsto (a, b)$$

is an isomorphism of additive groups.

The claim $\mathbb{C}^\times \cong \mathbb{R}^+ \times (\mathbb{C}, +)$ is false: $\mathbb{C}^\times$ contains nontrivial finite-order elements such as -1, whereas neither $\mathbb{R}^+$ under multiplication nor $\mathbb{C}$ under addition has nontrivial torsion. The correct polar decomposition is

$$\mathbb{C}^\times \cong \mathbb{R}^+ \times S^1, qquad z \longmapsto (|z|, z/|z|).$$

Useful invariants for disproving isomorphism include cardinality, element orders, divisibility, cyclicity, and existence of torsion. Thus:

- $(\mathbb{Z}, +)$ is not isomorphic to $(\mathbb{Q}, +)$ because $\mathbb{Z}$ is cyclic and $\mathbb{Q}$ is not;
- $(\mathbb{Q}, +)$ is not isomorphic to $(\mathbb{R}, +)$ because they have different cardinalities;
- $(\mathbb{R}, +)$ is not isomorphic to $(\mathbb{R}^\times, \cdot)$ because $\mathbb{R}^\times$ has the nonidentity element -1 of order 2, while $(\mathbb{R}, +)$ has no nonzero finite-order element;
- $(\mathbb{R}^\times, \cdot)$ is not isomorphic to $(\mathbb{Q}^\times, \cdot)$ because their cardinalities differ.

The group $\mathbb{Z}$ is isomorphic to its proper subgroup $2\mathbb{Z}$ via $n \mapsto 2n$.

11.11 The opposite group

For a group G, define G^{op} on the same underlying set by

$$a * b = ba.$$

Then $(a * b) * c = cba = a * (b * c)$, the identity is still e, and the inverse of a is still a^{-1}. Hence G^{op} is a group.

The inversion map

$$\iota : G \longrightarrow G^{\mathrm{op}}, \qquad a \longmapsto a^{-1}$$

is an isomorphism because

$$\iota(ab) = b^{-1}a^{-1} = a^{-1} * b^{-1} = \iota(a) * \iota(b),$$

and inversion is bijective.

11.12 A finite-group inverse-pairing argument

Suppose a finite group G has exactly one nonidentity element t satisfying $t^2 = e$. Elements other than e, t occur in distinct inverse pairs $\{g, g^{-1}\}$. Choose an ordering of the elements that places each pair next to one another. Then

$$\prod_{g \in G} g = e\, t \prod_{\{g, g^{-1}\}} g g^{-1} = t.$$

In a nonabelian group the notation $\prod_{g \in G} g$ requires an ordering; the displayed conclusion uses the permitted ordering in which inverse pairs are adjacent. The essential method is to pair every element with its inverse and observe that only the unique involution remains unpaired.

11.13 A modified product on a monoid

Let $(M, \cdot, 1)$ be a monoid and fix $m \in M$. Define

$$a *_m b = amb.$$

Then

$$(a *_m b) *_m c = (amb)mc = ambmc = am(bmc) = a *_m (b *_m c),$$

so this is a semigroup.

It has an identity u exactly when m is a unit in the original monoid. If u is a new identity, setting $a = 1$ in $u *_m a = a$ and $a *_m u = a$ gives

$$um = 1, qquadmu = 1,$$

so $u = m^{-1}$. Conversely, if m^{-1} exists, then $m^{-1} *_m a = a$ and $a *_m m^{-1} = a$.

11.14 Short manipulation questions

If $xyz^{-1}w = e$, preserve the order of factors:

$$xyz^{-1}w = e \implies yz^{-1}w = x^{-1} \implies yz^{-1} = x^{-1}w^{-1} \implies \boxed{y = x^{-1}w^{-1}z}.$$

If $xyz = e$, then

$$yzx = x^{-1}(xyz)x = e,$$

so cyclically rotating the factors preserves the identity. But $yxz = e$ need not follow: since $z = y^{-1}x^{-1}$, the expression $yxz = yxy^{-1}x^{-1}$ is a commutator and can be nontrivial in a nonabelian group.

Checkpoint: Group theory

Be able to do the following without notes:

1. apply the subgroup test, including to conjugate subgroups;
2. compute orders and generated subgroups in $\mathbb{Z}_n$;
3. classify a homomorphism from a cyclic group by the image of its generator;
4. reject a proposed quotient formula when it is not well-defined;
5. prove the inverse and square map criteria for commutativity;
6. use element order, divisibility, cyclicity, or cardinality to disprove an isomorphism;
7. verify product, opposite, and pointwise group axioms;
8. solve short noncommutative equations without changing factor order.

11.15 True/False drill

Decide each statement before reading the answers.

1. Every commutative magma is a semigroup.
2. Every finite nonempty subset of a group is a subgroup.
3. The image of a group homomorphism is a subgroup.
4. If $f : G \to H$ is a homomorphism, then $f(e_G) = e_H$.
5. Every subgroup of a cyclic group is cyclic.
6. The union of two subgroups is always a subgroup.
7. For every function f, upper star preserves complements.
8. For every function f, lower star preserves intersections.
9. If f^* is injective, then f is surjective.
10. If $g^2 = e$ for every $g \in G$, then G is abelian.

Answers. 1 false (associativity is separate); 2 false; 3 true; 4 true; 5 true; 6 false unless one subgroup contains the other; 7 true; 8 false without injectivity; 9 true; 10 true.

Chapter 12

Products and universal properties

The product $A \times B$ has projections p_1, p_2. Given $f : T \to A$, $g : T \to B$, there is a unique $\langle f, g \rangle : T \to A \times B$, $t \mapsto (f(t), g(t))$, with the given coordinates.

Intuition

To give a map into a product is exactly to give its coordinate maps. A universal property describes an object by its uniquely determined interactions.

Suppose $(a_1, b_1) \sim (a_2, b_2)$ iff $a_1 \sim_A a_2$ and $b_1 \sim_B b_2$. The coordinate maps are constant on classes, so quotient factorisation gives maps to $A/\!\sim_A$ and $B/\!\sim_B$. Pairing them gives

$$\Phi : (A \times B)/\!\sim \longrightarrow (A/\!\sim_A) \times (B/\!\sim_B), \qquad [(a, b)] \mapsto ([a], [b]).$$

It is surjective since $([a], [b]) = \Phi([(a, b)])$. If outputs agree, $a \sim_A a'$ and $b \sim_B b'$, so source classes agree; thus it is injective.

Coursework links

This is Assignment 3, Problem 10. The sheet *products and co products* provides more examples; coproducts are developed in the free-constructions chapter.

Chapter 13

Operations on quotients

To define $[a] \star [b] = [a * b]$, prove compatibility:

$$a \sim a',\ b \sim b' \implies a * b \sim a' * b'.$$

13.1 $\mathbb{Z}/n\mathbb{Z}$

The division algorithm gives $\mathbb{Z}/n\mathbb{Z} = \{[0], [1], \ldots, [n-1]\}$. Define $[a] +_n [b] = [a+b]$. If $a \equiv a'$ and $b \equiv b'$ modulo n, then n divides $(a+b) - (a'+b')$, proving well-definedness. Identity is $[0]$, inverse is $[-a]$, and associativity descends from $\mathbb{Z}$.

If $\pi : \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$ must be a magma morphism, then necessarily $[a] +_n [b] = \pi(a+b) = [a+b]$; hence the law is unique.

13.2 $\mathbb{Q}/\mathbb{Z}$

For $r \sim s \iff r - s \in \mathbb{Z}$, define $[r] + [s] = [r+s]$. The same subtraction calculation proves well-definedness; identity is $[0]$ and inverse $[-r]$. Every class has a representative in $[0, 1)$.

Practice 13.1. Find the order of $[2/5]$ in $\mathbb{Q}/\mathbb{Z}$.

Solution

$k[2/5] = [0]$ iff $2k/5 \in \mathbb{Z}$; the least positive k is 5.

Chapter 14

Free constructions: trees, words, and groups

Intuition

A free object is built from symbols with only the equations required by its type. Free magmas remember parentheses; free monoids forget parentheses but retain order; free commutative monoids also forget order; free groups additionally permit inverses. Each step is a quotient or enlargement with a precise universal property.

14.1 Free magmas: parenthesized binary trees

Let S be a set. A *formal binary tree with leaves in S* is either one leaf $s \in S$, or an ordered pair (u, v) of two smaller such trees. Write $L(S)$ for all finite trees. Distinguish leaves from pairs formally, even if S itself contains ordered pairs. Define $u \cdot v = (u, v)$. Then $L(S)$ is a magma, and the inclusion $i : S \to L(S)$ sends s to its one-leaf tree.

Theorem 14.1 (free magma)**.** *For every magma M and map $f : S \to M$, there is a unique magma morphism $\widehat{f} : L(S) \to M$ with $\widehat{f}(i(s)) = f(s)$.*

Proof. Define recursively $\widehat{f}(s) = f(s)$ on leaves and $\widehat{f}((u, v)) = \widehat{f}(u)\widehat{f}(v)$ on pairs. Every tree has a unique leaf-or-pair decomposition, so this defines exactly one value. The second rule is the morphism law. Any morphism agreeing with f on leaves is forced by that rule at each pair, hence induction on the number of leaves proves uniqueness. □

Practice 14.2. Are $((a, b), c)$ and $(a, (b, c))$ equal in the free magma? What happens under a morphism into a semigroup?

Solution

They are different trees: the left child of the first is a pair, while the left child of the second is the leaf a. Their images are $(f(a)f(b))f(c)$ and $f(a)(f(b)f(c))$, equal in a semigroup by associativity.

The smallest submagma of $L(S)$ containing $i(S)$ is all of $L(S)$, proved recursively. Any pair (P, p) with the same universal property is uniquely isomorphic to $(L(S), i)$: obtain maps in both directions by the property, then uniqueness forces their composites to be identities. In particular a magma morphism out of $L(S)$ is determined by its values on leaves. The free magma is generally neither associative nor commutative. If S is empty, $L(S)$ is empty under the convention allowing empty magmas; if the category requires nonempty magmas, the free object on $\varnothing$ does not exist there.

14.2 Free semigroups and free monoids: strings

Let S^+ be all nonempty finite strings $(s_1, \dots, s_n)$ of letters in S, and let $S^* = S^+ \cup \{()\}$ include the empty string. Concatenation is associative, and () is its identity. Thus S^+ is the free semigroup and S^* the free monoid on S.

Theorem 14.3 (free monoid)**.** *For any monoid M and map $f : S \to M$, exactly one monoid morphism $\widehat{f} : S^* \to M$ extends f, namely*

$$\widehat{f}(()) = e_M, \qquad \widehat{f}(s_1, \dots, s_n) = f(s_1) \cdots f(s_n).$$

Proof. The empty string must map to e_M. Every nonempty string is a concatenation of its one-letter strings, so the displayed formula is forced. Associativity makes it unambiguous and shows concatenation maps to multiplication. □

Practice 14.4. Let $S = \{a, b\}$ and send $a \mapsto 2$, $b \mapsto 3$ into $(\mathbb{N}, \cdot)$. Compute the image of the word *abba* and the empty word.

Solution

$\widehat{f}(abba) = 2 \cdot 3 \cdot 3 \cdot 2 = 36$, while the empty word maps to the multiplicative identity 1.

The free semigroup S^+ is the subsemigroup generated by the letters; S^* is the submonoid generated by them. For a one-letter alphabet, $S^* \cong (\mathbb{N}_0, +)$ by word length. For a zero-letter alphabet it is the one-element monoid. A free monoid also arises by quotienting the free magma after imposing associativity and adjoining the empty word; the order of letters remains visible.

14.3 Free commutative monoids and free abelian groups

Let $\mathbb{N}_0^{(S)}$ be the set of finitely supported functions $m : S \to \mathbb{N}_0$, with pointwise addition. A word maps to its *multiplicity function*: $m(s)$ counts occurrences of s. This is a surjective monoid morphism $S^* \to \mathbb{N}_0^{(S)}$ whose kernel congruence is generated by all swaps $st \sim ts$. Hence $\mathbb{N}_0^{(S)}$ is the free commutative monoid on S. Similarly, $\mathbb{Z}^{(S)}$ consists of finitely supported integer-valued functions. Under addition it is the free abelian group (or free $\mathbb{Z}$-module) on S. Write e_s for the function that is 1 at s and 0 elsewhere. Every element is uniquely a finite sum $\sum_s n_s e_s$. For any abelian group A and map $f : S \to A$, the unique homomorphism extending f is $\sum_s n_s e_s \mapsto \sum_s n_s f(s)$.

Practice 14.5. Identify $\mathbb{N}_0^{(\{a,b\})}$ and $\mathbb{Z}^{(\{a,b\})}$ with familiar structures. Where does *abba* go?

Solution

They are respectively $\mathbb{N}_0^2$ as a commutative monoid and $\mathbb{Z}^2$ as an abelian group, using coordinates $(m(a), m(b))$. The word *abba* maps to $(2, 2)$.

If C is the smallest group congruence on a free group $F(S)$ identifying xy with yx for every x, y, then $F(S)/C \cong \mathbb{Z}^{(S)}$. Explicitly, send each free generator to e_s; its kernel is the subgroup generated by commutators. This is the free-abelian reformulation of abelianization used later.

14.4 Free groups, reduced words, and presentations

The preceding chapter introduced reduced words; here is their formal construction. Formally take letters $S \sqcup S^{-1}$, where each s^{-1} is a new formal symbol. Delete adjacent ss^{-1} or $s^{-1}s$ until

reduced. The result is unique; concatenation followed by reduction defines the free group $F(S)$. The empty word is identity and reversing a word while inverting each letter gives its inverse. The free property says any map $S \to G$ extends uniquely to a group morphism $F(S) \to G$. For a set R of words, $\langle S \mid R \rangle$ is $F(S)$ modulo the smallest *normal* subgroup containing R. Equivalently it is the group generated by S subject to each $r \in R$ equalling e. A map $S \to H$ extends to $\langle S \mid R \rangle$ iff every relator evaluates to e_H; the extension is then unique. Relations must hold under conjugation too, which is why the normal closure rather than merely the subgroup generated by R is used.

Practice 14.6. Show $\langle a, b \mid ab = e \rangle \cong \mathbb{Z}$.

Solution

The relation forces $b = a^{-1}$, so every word reduces to a power of a. Send $a \mapsto 1$, $b \mapsto -1$ into $(\mathbb{Z}, +)$; the relator maps to 0, so this defines a homomorphism. Conversely send $1 \mapsto a$ from $\mathbb{Z}$ to the presented group. These maps fix the generators in both directions, hence are inverse.

14.5 Products, coproducts, and their different underlying sets

For groups $(G_i)_{i \in I}$, the Cartesian product $\prod_i G_i$ is a group under coordinatewise multiplication. A map $H \to \prod_i G_i$ is a homomorphism exactly when each coordinate map is one. This is the product universal property. The *coproduct* $*_{i \in I} G_i$, called the free product, has homomorphisms $\iota_i : G_i \to *_{i \in I} G_i$ such that any family $f_i : G_i \to H$ extends uniquely to $f : *_{i \in I} G_i \to H$. One construction starts with $F(\bigsqcup_i G_i)$ and imposes relations $[g][h] = [gh]$ for g, h from the same factor and $[e_i] = e$. It leaves words alternating between distinct factors unreduced. For example $C_2 * C_3$ is infinite, whereas $C_2 \times C_3$ has six elements. The forgetful functor to sets does not turn a group coproduct into a disjoint union of underlying sets. In the category of abelian groups, the coproduct is the *direct sum* $\bigoplus_i A_i$: the subgroup of $\prod_i A_i$ consisting of tuples with finite support. For finitely many factors, direct sum and direct product coincide as groups; for infinitely many, they differ in general. Since $\mathbb{Z}$ is the free abelian group on one generator, $\bigoplus_{s \in S} \mathbb{Z} \cong \mathbb{Z}^{(S)}$ is the free abelian group on S.

Practice 14.7. Why is $C_2 * C_3$ not the same as $C_2 \times C_3$?

Solution

In the product, elements from the two factors commute: $(g, e)(e, h) = (e, h)(g, e)$. In the free product no such cross-factor relation is imposed; alternating words such as $ghgh$ remain distinct after reduction. Thus the free product is infinite and noncommutative, while $C_2 \times C_3$ is finite and abelian.

Checkpoint: Lectures 4–6

Know what each free construction remembers: tree shape, word order, letter counts, or signed letter counts. Prove a universal property by constructing the forced map and then uniqueness. Distinguish product from coproduct and direct product from direct sum.

Chapter 15

Free groups and presentations

15.1 Reduced words

For symbols A, add formal inverses. A word is reduced if no adjacent aa^{-1} or $a^{-1}a$ occurs. Multiply by concatenating and cancelling inverse pairs. The empty word is identity. This is the free group $F(A)$.

Intuition

Free means no accidental equations: two reduced words represent the same element only when literally identical.

Theorem 15.1 (free universal property)**.** *Every function $f : A \to G$ extends uniquely to a homomorphism $\widehat{f} : F(A) \to G$.*

Proof. Send $a_1^{\epsilon_1} \cdots a_k^{\epsilon_k}$ to $f(a_1)^{\epsilon_1} \cdots f(a_k)^{\epsilon_k}$. Cancellation does not change the value. Values on generators force every word, giving uniqueness. □

15.2 The rank-three subgroup

Define $\epsilon : F(x, y) \to \mathbb{Z}/2\mathbb{Z}$ by $\epsilon(x) = \epsilon(y) = 1$. Its kernel K has index two. Using coset representatives $\{1, x\}$, Reidemeister–Schreier gives the free basis

$$x^2, \qquad yx^{-1}, \qquad xy.$$

A Nielsen move may replace a basis element by its product with another. Since $(yx^{-1})(xy) = y^2$, the triple $u = x^2, v = y^2, z = xy$ is also a free basis. Therefore $\langle u, v, z\rangle \cong F(u, v, z)$, solving Problem 13.

To justify the basis, either derive it from the Reidemeister–Schreier theorem or state the theorem and show the calculation. The absence of an obvious relation is not a proof.

15.3 Presentations and eliminating generators

$\langle x_1, \ldots, x_n \mid r_1, \ldots, r_m\rangle$ is the free group modulo the normal closure of the relators. A generator uniquely solved for by a relation can be eliminated (a Tietze transformation).

For $G = \langle x, y, z \mid yxyz^{-2} = e\rangle$, the relation is $x = y^{-1}z^2y^{-1}$. Eliminate x to obtain $G \cong \langle y, z \mid \rangle = F(y, z)$. Explicitly, the map back sends x to $y^{-1}z^2y^{-1}$ and fixes y, z; the relator becomes identity, and the two maps are inverse. This solves Problem 14.

15.4 Why $\langle x, y \mid xyx^{-1}y^{-1} \rangle$ is $\mathbb{Z} \times \mathbb{Z}$

Let

$$P = \langle x, y \mid xyx^{-1}y^{-1} = e \rangle .$$

The relator says $xy = yx$, so every word may be rearranged and reduced to $x^m y^n$ for integers m, n. Define

$$\Phi : \mathbb{Z} \times \mathbb{Z} \to P, qquad \Phi(m, n) = x^m y^n.$$

Because x, y commute,

$$\Phi(m, n)\Phi(r, s) = x^{m+r} y^{n+s} = \Phi(m + r, n + s),$$

so Φ is a surjective homomorphism.

Conversely, send $x \mapsto (1, 0)$ and $y \mapsto (0, 1)$. The relator maps to $(0, 0)$ because $\mathbb{Z}^2$ is abelian, so this defines a homomorphism

$$\Psi : P \to \mathbb{Z} \times \mathbb{Z}.$$

Now $\Psi\Phi(m, n) = (m, n)$, while $\Phi\Psi$ fixes both generators x, y and hence every element of P. Thus the maps are inverse and

$$P \cong \mathbb{Z} \times \mathbb{Z}.$$

Chapter 16

Categories, functors, and universal properties

16.1 From quivers to categories

Definition 16.1. A *quiver* consists of a set Q_0 of vertices, a set Q_1 of arrows, and source and target maps $s, t : Q_1 \to Q_0$. A quiver morphism is a pair of maps on vertices and arrows commuting with source and target.

A directed graph may have parallel arrows, loops, or no arrows. The quiver with vertices A, B and one arrow $A \to B$ is not yet a category: a category must also have identities and composites. Its *free category* has vertices as objects and all finite directed paths as arrows, including length-zero paths id_A. Composition is path concatenation. Associativity follows from associativity of concatenation.

Definition 16.2. A *category* $\mathcal{C}$ has objects, sets $\mathrm{Hom}_{\mathcal{C}}(A, B)$ of arrows, associative composition $\mathrm{Hom}(B, C) \times \mathrm{Hom}(A, B) \to \mathrm{Hom}(A, C)$, and an identity id_A for each A with $f \,\mathrm{id}_A = f = \mathrm{id}_B \, f$. An arrow $A \to B$ is an *isomorphism* if it has a two-sided inverse $B \to A$.

Intuition

A category records objects and legal structure-preserving maps. The identity and composition axioms let us reason about maps without looking inside the objects.

Examples are **Set** (sets/functions), **Rel** (sets/relations), magmas/magma morphisms, semigroups, unital magmas, monoids, commutative monoids, groups, and abelian groups (also called $\mathbb{Z}$-modules). In **Rel** a relation $R : A \to B$ is a subset of $A \times B$; composition is $a(S \circ R)c$ iff some b satisfies aRb and bSc, and identity is equality. Every monoid M can be regarded as a one-object category with arrows M and composition its multiplication. A group is a one-object category in which every arrow is invertible. A partially ordered set is a category with one arrow $a \to b$ exactly when $a \leq b$; a preorder also works.

Practice 16.3. In a poset-category, when are two objects isomorphic? In **Rel**, check that the diagonal equality relation is an identity.

Solution

An isomorphism $a \to b$ requires both $a \leq b$ and $b \leq a$; antisymmetry gives $a = b$. For $R \subseteq A \times B$, $(a, b) \in R \circ \Delta_A$ iff some a' has $a = a'$ and $(a', b) \in R$, equivalent to $(a, b) \in R$. Similarly $\Delta_B \circ R = R$.

Identities and inverses in any category are unique. If g and h are both inverses of f, then $g = g(fh) = (gf)h = h$. Every $\mathrm{End}_{\mathcal{C}}(A) = \mathrm{Hom}_{\mathcal{C}}(A, A)$ is a monoid under composition; its units form the automorphism group $\mathrm{Aut}_{\mathcal{C}}(A)$.

16.2 Pointed sets form a category

A pointed set is (S, s_0). A morphism $(S, s_0) \to (T, t_0)$ is a function $f : S \to T$ with $f(s_0) = t_0$. The identity preserves the basepoint. If $f(s_0) = t_0$ and $g(t_0) = u_0$, then $(g \circ f)(s_0) = u_0$, so morphisms are closed under composition. Associativity and identity laws are inherited from ordinary functions. Hence pointed sets form a category.

16.3 Monomorphisms and epimorphisms

Definition 16.4. An arrow $f : A \to B$ is a *monomorphism* if $fg_1 = fg_2$ implies $g_1 = g_2$ for all parallel arrows $g_1, g_2 : C \to A$. It is an *epimorphism* if $h_1 f = h_2 f$ implies $h_1 = h_2$ for all parallel arrows $h_1, h_2 : B \to D$.

These are categorical left and right cancellation. In **Set**, monomorphisms are exactly injections and epimorphisms exactly surjections. For the converse mono test, if $f(a) = f(a')$, use two maps from a singleton selecting a, a'. For the converse epi test, if $b \notin f(A)$, choose two maps $B \to \{0, 1\}$ that differ only at b. In other categories, underlying-set injection/surjection need not characterize mono/epi. Every isomorphism is both, but the converse can fail (for instance $\mathbb{Z} \hookrightarrow \mathbb{Q}$ is both mono and epi in the category of rings with identity, but not an isomorphism).

Practice 16.5. Show directly that a surjective group morphism is an epimorphism in **Grp**.

Solution

If $h_1 f = h_2 f$ and $b \in B$, choose a with $f(a) = b$. Then $h_1(b) = h_1 f(a) = h_2 f(a) = h_2(b)$. Thus $h_1 = h_2$.

Common trap

A handwritten example in Lecture 7 labels the inclusion $(\mathbb{Z}, +) \hookrightarrow (\mathbb{Q}, +)$ an epimorphism in commutative monoids. As stated this is false: the quotient map $q : \mathbb{Q} \to \mathbb{Q}/\mathbb{Z}$ and the zero map are distinct monoid morphisms but agree on every integer. Do not use this example as a proof. In the category of *unital rings*, the same underlying inclusion is an epimorphism: two ring maps out of $\mathbb{Q}$ agreeing on $\mathbb{Z}$ agree on every fraction because they preserve inverses.

16.4 Opposite categories and equivalences

For any category $\mathcal{C}$, the *opposite category* $\mathcal{C}^{\mathrm{op}}$ has the same objects, with $\mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(A, B) = \mathrm{Hom}_{\mathcal{C}}(B, A)$; composition is reversed. A contravariant functor $\mathcal{C} \to \mathcal{D}$ is equivalently a covariant functor $\mathcal{C}^{\mathrm{op}} \to \mathcal{D}$. Products in $\mathcal{C}$ become coproducts in $\mathcal{C}^{\mathrm{op}}$, and initial objects become terminal objects. An *equivalence of categories* consists of functors $F : \mathcal{C} \to \mathcal{D}$, $G : \mathcal{D} \to \mathcal{C}$ with natural isomorphisms $GF \cong \mathrm{id}_{\mathcal{C}}$ and $FG \cong \mathrm{id}_{\mathcal{D}}$. This is weaker than an isomorphism of categories, which requires literal equality of the composites. For example, finite-dimensional vector spaces over a fixed field form a category equivalent to the category whose objects are natural numbers n and whose arrows $m \to n$ are $n \times m$ matrices: choosing a basis supplies the comparison, but the choices are not canonical equalities of objects.

Practice 16.6. What does the opposite category of a poset P represent?

Solution

It represents the reversed order: there is an arrow $a \to b$ in P^{op} exactly when $b \leq a$ in P.

16.5 Functors and the forgetful/free constructions

Definition 16.7. A *covariant functor* $F : \mathcal{C} \to \mathcal{D}$ assigns an object $F(A)$ to every A and an arrow $F(f) : F(A) \to F(B)$ to every $f : A \to B$, preserving $F(\mathrm{id}_A) = \mathrm{id}_{F(A)}$ and $F(gf) = F(g)F(f)$. A *contravariant functor* reverses arrow directions and satisfies $F(gf) = F(f)F(g)$.

Intuition

A functor is a structure-preserving map between entire categories. A homomorphism preserves multiplication; a functor preserves composition.

The identity functor does nothing. Functors compose. Every functor sends isomorphisms to isomorphisms because $F(f)F(f^{-1}) = F(\mathrm{id}) = \mathrm{id}$. The forgetful functor $U : \mathbf{Grp} \to \mathbf{Set}$ sends a group to its underlying set and a homomorphism to its underlying function. Free magma, free monoid, free group, and free abelian group constructions are functors from **Set** to their categories: a map $u : S \to T$ acts by replacing each generator s with $u(s)$, and the free universal property gives the unique induced morphism. The constant functor sends all objects to one fixed object and all arrows to its identity.

Practice 16.8. Why is $F(u \circ v) = F(u) \circ F(v)$ for a free-group functor?

Solution

Both sides are homomorphisms from $F(R)$ to $F(T)$ and agree on each free generator r, sending it to the generator $(u \circ v)(r)$. Uniqueness in the free universal property makes them equal.

16.6 Natural transformations and functor categories

Definition 16.9. For functors $F, G : \mathcal{C} \to \mathcal{D}$, a *natural transformation* $\eta : F \Rightarrow G$ gives an arrow $\eta_A : F(A) \to G(A)$ for every object A, such that for every $f : A \to B$,

$$G(f)\eta_A = \eta_B F(f).$$

It is a *natural isomorphism* if every η_A is an isomorphism.

Intuition

The square says it does not matter whether we first change the object using f and then apply η, or apply η first. "Natural" means no arbitrary choice of coordinates is hidden.

For group G, the map $\eta_G : G \to G_{\mathrm{ab}} = G/[G, G]$ is natural in G: a homomorphism $f : G \to H$ induces f_{ab} and $f_{\mathrm{ab}}\eta_G = \eta_H f$. Natural transformations compose componentwise: $(\theta\eta)_A = \theta_A\eta_A$; their identity has components $\mathrm{id}_{F(A)}$. Thus functors $\mathcal{C} \to \mathcal{D}$ and natural transformations form a *functor category* $[\mathcal{C}, \mathcal{D}]$ (with the usual size qualification). There is also a category **Cat** whose objects are (small) categories and arrows are functors; composition of functors and identity functors satisfy the category axioms. The objects of $[\mathcal{C}, \mathcal{D}]$ are functors, whereas its arrows are natural transformations. These are two different levels of structure.

Practice 16.10. For a fixed set K, let $F(S) = S \times K$ and $G(S) = S$. Show the first-coordinate projections $\eta_S : S \times K \to S$ are natural.

Solution

For $u : S \to T$, $F(u)(s,k) = (u(s),k)$. Then $u\eta_S(s,k) = u(s) = \eta_T F(u)(s,k)$, so the naturality square commutes.

16.7 Representable functors and the Yoneda idea

Fix an object X of $\mathcal{C}$. The covariant functor $h_X = \mathrm{Hom}_{\mathcal{C}}(X,-)$ sends $f : A \to B$ to postcomposition $u \mapsto fu$. The contravariant functor $h^X = \mathrm{Hom}_{\mathcal{C}}(-,X)$ sends $f : A \to B$ to precomposition $v \mapsto vf$. A functor naturally isomorphic to one of these is called *representable* (with variance specified).

Theorem 16.11 (Yoneda, covariant form)**.** *For $T : \mathcal{C} \to \mathbf{Set}$, natural transformations $h_X \Rightarrow T$ correspond bijectively to elements of $T(X)$ by $\eta \mapsto \eta_X(\mathrm{id}_X)$. Given $a \in T(X)$, the inverse sends $u : X \to A$ to $T(u)(a)$.*

Proof. For natural η and $u : X \to A$, naturality for u gives $\eta_A(u) = T(u)(\eta_X(\mathrm{id}_X))$. This proves uniqueness. Conversely the displayed formula is natural since for $v : A \to B$, $T(v)T(u)(a) = T(vu)(a)$. □

Practice 16.12. What is a natural transformation $\mathrm{Hom}(X,-) \Rightarrow \mathrm{Hom}(Y,-)$ determined by?

Solution

Yoneda identifies it with an element of $\mathrm{Hom}(Y,X)$. Such an arrow $a : Y \to X$ gives the component $u : X \to A \mapsto u \circ a : Y \to A$. Note the reversed direction $Y \to X$.

The dual Yoneda statement applies to contravariant representables. Representable functors preserve limits: for example $\mathrm{Hom}(X, A \times B) \cong \mathrm{Hom}(X,A) \times \mathrm{Hom}(X,B)$ naturally, and $\mathrm{Hom}(X,1)$ has exactly one element when 1 is terminal. The word "naturally" in Yoneda matters. The map $\eta \mapsto \eta_X(\mathrm{id}_X)$ commutes with changing either the representing object X or the target functor T. Thus Yoneda identifies an object by all of its interactions with other objects, not merely by a count of elements.

16.8 Initial, terminal, products, and coproducts

An object 0 is *initial* if each $\mathrm{Hom}(0,A)$ has exactly one element. An object 1 is *terminal* if each $\mathrm{Hom}(A,1)$ has exactly one element. Either is unique up to unique isomorphism. In **Set**, $\varnothing$ is initial and a singleton terminal; in **Grp**, the trivial group is both. In **Mon** the one-element monoid is both. In the category of pointed sets, the one-pointed set is both. In a poset-category, initial and terminal objects are least and greatest elements. A *product* $P = \prod_i A_i$ has projections $p_i : P \to A_i$ such that every family $f_i : T \to A_i$ has exactly one pairing $\langle f_i \rangle : T \to P$ with $p_i \langle f_i \rangle = f_i$. A *coproduct* reverses the arrows: inclusions $j_i : A_i \to C$ and one unique map $C \to T$ for every family $A_i \to T$. In **Set** these are Cartesian product and disjoint union. In **Grp** the product is componentwise Cartesian product, while the coproduct is free product. In **Ab** the coproduct is direct sum. Every product or coproduct is unique up to unique isomorphism compatible with its structure maps.

Practice 16.13. Find the coproduct of pointed sets (A,a_0) and (B,b_0).

Solution

Take the disjoint union $A \sqcup B$ and identify the two basepoints. Denote the common point by $*$. Any two basepoint-preserving maps $A \to T$, $B \to T$ agree at the identified basepoints (both send them to t_0), so they define exactly one basepoint-preserving map from this quotient to T.

16.9 Cartesian and cocartesian squares

Given arrows $A \xrightarrow{f} C \xleftarrow{g} B$, their *pullback* or Cartesian square is a universal object with maps to A, B whose composites into C agree. In **Set** it is

$$A \times_C B = \{(a, b) \in A \times B : f(a) = g(b)\}.$$

Given $C \xrightarrow{f} A$ and $C \xrightarrow{g} B$, the *pushout* or cocartesian square is the dual universal object. In **Set** it is $(A \sqcup B)/\sim$, where $f(c) \sim g(c)$ for every $c \in C$, with the smallest equivalence relation containing these pairs. In a poset regarded as a category, a pullback of $a \leq c$, $b \leq c$ is their greatest lower bound $a \wedge b$; a pushout of $c \leq a$, $c \leq b$ is their least upper bound $a \vee b$, when these exist. In $\mathcal{P}(S)$ they are intersection and union.

Practice 16.14. Let $f : \{1, 2\} \to \{a, b\}$ send both inputs to a, and $g : \{u, v\} \to \{a, b\}$ send $u \mapsto a$, $v \mapsto b$. Find the pullback.

Solution

The pairs satisfying $f(x) = g(y)$ are exactly $(1, u)$ and $(2, u)$. The pair with v is excluded because $g(v) = b$.

16.10 Adjunctions and the naturality requirement

Definition 16.15. Functors $F : \mathcal{C} \to \mathcal{D}$ and $U : \mathcal{D} \to \mathcal{C}$ form an *adjunction*, written $F \dashv U$, when there are bijections

$$\mathrm{Hom}_{\mathcal{D}}(F(X), Y) \;\cong\; \mathrm{Hom}_{\mathcal{C}}(X, U(Y))$$

natural in both X and Y. Thus an arrow out of $F(X)$ is the same data as an arrow out of X into $U(Y)$, compatibly with maps in both variables.

The free group functor is left adjoint to the forgetful functor **Grp** $\to$ **Set**; the bijection is restriction of a homomorphism $F(S) \to G$ to the generators S. Free monoid, free magma, and free abelian group functors give analogous adjunctions. The abelianization adjunction appears in Lecture 13 below. For a set map $f : X \to Y$, the power-set adjunctions $f_* \dashv f^* \dashv f_!$ from the earlier chapter are order-theoretic examples: $f_*(A) \subseteq B \iff A \subseteq f^*(B)$ and $f^*(B) \subseteq A \iff B \subseteq f_!(A)$.

Practice 16.16. Explain why a mere bijection of two Hom-sets for one choice of X, Y is not enough to call functors adjoint.

Solution

An adjunction requires a bijection for *every* X, Y, and these bijections must commute with precomposition in X and postcomposition in Y. Otherwise unrelated, arbitrarily chosen bijections of equal-sized sets would falsely count as an adjunction.

Checkpoint: Lectures 7–8

Define a category, functor, natural transformation, mono, and epi from their equations. Compute initial/terminal objects, products/coproducts, and a pullback in examples. For any proposed universal property, construct the forced arrow, prove it respects structure, then prove uniqueness.

Chapter 17

Actions, orbits, cosets, and conjugation

17.1 Monoid actions and equivariant maps

Definition 17.1. A *left action* of a monoid M on a set S is a map $M \times S \to S$, $(m, s) \mapsto m \cdot s$, such that $e \cdot s = s$ and $(mn) \cdot s = m \cdot (n \cdot s)$. The pair is an M-set. An *equivariant map* $h : S \to T$ of M-sets satisfies $h(m \cdot s) = m \cdot h(s)$ for all m, s.

Intuition

An action gives a concrete transformation of S for each abstract monoid element. The multiplication rule says the transformation for mn is the composite transformation.

Writing $\rho(m)(s) = m \cdot s$, the action laws say precisely that $\rho : M \to \mathrm{End}_{\mathbf{Set}}(S)$ is a monoid morphism, where $\mathrm{End}(S)$ is all self-maps under composition. Conversely every such morphism defines an action. If $M = G$ is a group, each $\rho(g)$ is invertible, with inverse $\rho(g^{-1})$, so ρ lands in $\mathrm{Sym}(S)$. An equivariant map is exactly an arrow between actions; actions and equivariant maps form a category M-**Set**. The regular left action of G on itself is $g \cdot x = gx$. If $\phi : H \to G$ is a group morphism and S a G-set, then $h \cdot s = \phi(h) \cdot s$ defines the *restricted* H-action.

Practice 17.2. Let G act on itself by left multiplication. Show that $h_a(g) = ga$ is equivariant for every fixed $a \in G$.

Solution

$h_a(xg) = (xg)a = x(ga) = x \cdot h_a(g)$ by associativity. Left multiplication $g \mapsto ag$ would not generally be equivariant unless a were central.

17.2 Orbits and invariant subsets

For a group action, define the orbit of s by $G \cdot s = \{g \cdot s : g \in G\}$. The relation $s \sim t \iff t = g \cdot s$ for some g is an equivalence relation: use e, g^{-1}, and multiplication for reflexivity, symmetry, transitivity. Thus the orbits partition S, and the set of orbits is denoted $G\backslash S$ (for a left action). Two orbits that meet are equal. A subset $T \subseteq S$ is G-invariant if $gT \subseteq T$ for all g; for groups this implies $gT = T$ and means T is a union of whole orbits. The orbit $G \cdot s$ is the smallest invariant subset containing s.

Practice 17.3. Let $\mathbb{Z}$ act on $\mathbb{Z}$ by $n \cdot x = x + 2n$. Describe the orbits.

Solution

The orbit of x is $x + 2\mathbb{Z}$. There are two orbits: even and odd integers.

For a general monoid action, the reachability relation $t = m \cdot s$ need not be symmetric, so one should not claim it is an equivalence relation without group inverses.

17.3 Cosets, index, and the regular action

Let $H \leq G$. Left multiplication of H on G gives orbits $Hg = \{hg : h \in H\}$, the *right cosets* in the convention where the subgroup symbol is on the left; right multiplication gives gH, the *left cosets*. The set $G/H = \{gH : g \in G\}$ is the set of left cosets; $H\backslash G = \{Hg : g \in G\}$ is the set of right cosets. Each is a partition of G. For example

$$g_1H = g_2H \iff g_2^{-1}g_1 \in H, \qquad Hg_1 = Hg_2 \iff g_1g_2^{-1} \in H.$$

The *index* $[G : H]$ is the number of left cosets, finite or infinite. Inversion $gH \mapsto Hg^{-1}$ is a bijection between left and right coset sets, so they have the same number. If G is finite, each coset has $|H|$ elements via $h \mapsto gh$, whence $|G| = [G : H]|H|$ (Lagrange's theorem).

Practice 17.4. List the cosets of $3\mathbb{Z}$ in $(\mathbb{Z}, +)$ and compute the index.

Solution

The cosets are $3\mathbb{Z}$, $1 + 3\mathbb{Z}$, and $2 + 3\mathbb{Z}$, partitioning $\mathbb{Z}$ by residue modulo 3. Thus $[\mathbb{Z} : 3\mathbb{Z}] = 3$.

Common trap

G/H is always a *set* of cosets. It is a group under $gH \cdot kH = gkH$ only when H is normal; do not assume quotient multiplication for arbitrary H.

17.4 Stabilizers and orbit–stabilizer

Definition 17.5. For s in a G-set, its *stabilizer* is $G_s = \{g \in G : g \cdot s = s\}$.

It is a subgroup: e fixes s; if g, h fix s, then $gh^{-1} \cdot s = g \cdot (h^{-1} \cdot s) = s$ since h^{-1} also fixes s. If $t = g \cdot s$, then $G_t = gG_sg^{-1}$, so stabilizers along one orbit are conjugate.

Theorem 17.6 (orbit–stabilizer)**.** *The map $G/G_s \to G \cdot s$, $gG_s \mapsto g \cdot s$, is a well-defined bijection. Consequently, for finite G, $|G \cdot s| = [G : G_s] = |G|/|G_s|$.*

Proof. We have $gG_s = hG_s$ iff $h^{-1}g \in G_s$, iff $(h^{-1}g) \cdot s = s$, iff $g \cdot s = h \cdot s$. This proves both well-definedness and injectivity; surjectivity is the definition of the orbit. The count follows from Lagrange's theorem. □

Practice 17.7. Let S_3 act on $\{1, 2, 3\}$ by permutations. Find $|G_1|$ and $|G \cdot 1|$.

Solution

The stabilizer of 1 consists of the identity and the transposition (23), so $|G_1| = 2$. The orbit is all three points, and $6/2 = 3$ confirms orbit–stabilizer.

More generally the size of an orbit divides $|G|$ when G is finite. Counting all disjoint orbits yields $|S| = \sum_{[s] \in G\backslash S} |G \cdot s|$ for finite S.

17.5 Conjugation, centralizers, and the centre

The *conjugation action* of G on itself is $g \cdot x = gxg^{-1}$. Its orbit is the conjugacy class of x and its stabilizer is the *centralizer* $C_G(x) = \{g : gx = xg\}$. Thus $|\text{class}(x)| = [G : C_G(x)]$ in a finite group. Conjugacy is an equivalence relation. A conjugate subgroup is $gHg^{-1} = \{ghg^{-1} : h \in H\}$, again a subgroup; conjugation by g is an isomorphism $H \to gHg^{-1}$. The map $\text{Int} : G \to \text{Aut}(G)$, $g \mapsto (x \mapsto gxg^{-1})$, is a homomorphism since $\text{Int}(gh) = \text{Int}(g)\,\text{Int}(h)$. Its kernel is the *centre*

$$Z(G) = \{g \in G : gx = xg \text{ for every } x \in G\}.$$

Equivalently $Z(G) = \bigcap_{x \in G} C_G(x)$. It is a normal subgroup: conjugation by any group element sends a central element to itself. The conjugacy classes of size one are exactly the elements of $Z(G)$.

Practice 17.8. Find $Z(S_3)$.

Solution

A central permutation must commute with both (12) and (23). Direct checking of the six permutations shows only the identity does. Equivalently, conjugating a transposition by a suitable permutation produces a different transposition, and the nonidentity 3-cycles also fail to commute with a transposition. Hence $Z(S_3) = \{e\}$.

17.6 The class equation and p-groups

For a finite group, split G into its conjugacy classes. The singleton classes are $Z(G)$; choose one representative x_i from each remaining class. Then

$$|G| = |Z(G)| + \sum_i [G : C_G(x_i)].$$

This is the *class equation*. More generally any finite G-set has $|S| = |S^G| + \sum_i [G : G_{s_i}]$, where S^G is the set of points fixed by all of G and the sum runs over nontrivial orbits.

Definition 17.9. A finite group of order p^n for some prime p and $n \geq 1$ is a p-group.

If a p-group acts on a finite set S, every nontrivial orbit has size divisible by p, because its size is a nontrivial divisor of p^n. Hence $|S| \equiv |S^G| \pmod p$. Apply this to conjugation on $S = G$: $|G| \equiv |Z(G)| \pmod p$. Since $e \in Z(G)$, the centre is nonempty; since $p \mid |G|$, we obtain $p \mid |Z(G)|$ and thus $|Z(G)| > 1$.

Practice 17.10. Why does every group of order p^2 have to be abelian?

Solution

Its centre is nontrivial, so $|Z(G)|$ is p or p^2. If it is p^2, then $Z(G) = G$. If it is p, then $G/Z(G)$ has order p and is cyclic. But if $G/Z(G) = \langle gZ(G)\rangle$, any $x, y \in G$ are $g^a z$ and $g^b w$ with z, w central, so they commute. Then G is abelian, forcing $Z(G) = G$, a contradiction. Hence only the first case occurs.

Checkpoint: Lectures 9–11

Translate actions to homomorphisms into $\text{Sym}(S)$; compute orbits and stabilizers; distinguish gH from Hg; prove orbit–stabilizer; use conjugation to identify centralizers and the

centre; apply the class equation to p-groups.

Chapter 18

Normal subgroups, quotients, and abelianization

18.1 Normality and the quotient group

Definition 18.1. A subgroup $N \leq G$ is *normal*, written $N \trianglelefteq G$, if $gNg^{-1} = N$ for every $g \in G$. Equivalently $gN = Ng$ for every g, or the left and right coset partitions coincide.

Proof. If $gNg^{-1} = N$, multiply on the right by g to get $gN = Ng$. Conversely multiply $gN = Ng$ on the right by g^{-1}. The equality of coset partitions is equivalent because the left coset gN and right coset Ng both contain g, so if the partitions coincide they must be the same block. □

Intuition

Normality is precisely the condition that multiplication of cosets does not depend on which representatives are chosen.

If $N \trianglelefteq G$, define $(gN)(hN) = ghN$. To prove well-definedness, write $g' = gn$ and $h' = hm$ with $n, m \in N$. Then $g'h' = gnhm = gh(h^{-1}nh)m$, and the parenthesized factors lie in N by normality; thus $g'h'N = ghN$. Identity is N, inverse of gN is $g^{-1}N$, and associativity descends from G. The projection $\pi : G \to G/N$ is a surjective group morphism with kernel N.

Practice 18.2. Show that the subgroup $\langle (12) \rangle$ is not normal in S_3, while $A_3 = \{e, (123), (132)\}$ is normal.

Solution

Conjugating (12) by (123) gives a different transposition, so its two-element subgroup is not preserved. A_3 has index two in S_3; there are only two left cosets and two right cosets, each partition with one block A_3 and one complementary block, so the cosets coincide. Thus $A_3 \trianglelefteq S_3$.

Arbitrary intersections of normal subgroups are normal. Every kernel of a group homomorphism is normal: for $n \in \ker f$, $f(gng^{-1}) = f(g)ef(g)^{-1} = e$. Conversely every normal subgroup is the kernel of its quotient projection. The centre $Z(G)$ and derived subgroup below are normal. A group is *simple* if it is nontrivial and its only normal subgroups are $\{e\}$ and G; the word *normal* cannot be omitted.

18.2 Universal property and the first isomorphism theorem

Let $N \trianglelefteq G$ and $f : G \to H$ be a group morphism. There is a unique group morphism $\bar{f} : G/N \to H$ with $f = \bar{f}\pi$ exactly when $N \subseteq \ker f$. Necessity follows because $\pi(n) = N$ for $n \in N$. For sufficiency define $\bar{f}(gN) = f(g)$; if $gN = hN$, then $h^{-1}g \in N \subseteq \ker f$, so $f(g) = f(h)$. The formula preserves multiplication, and surjectivity of π forces uniqueness. The induced map is injective iff $N = \ker f$ and is surjective iff f is surjective. In particular, when $N = \ker f$, it gives the *first isomorphism theorem*

$$G/\ker f \;\cong\; \operatorname{im} f, \qquad g \ker f \longmapsto f(g).$$

Practice 18.3. Use the theorem to show $\mathbb{Z}/3\mathbb{Z} \cong \{1, \omega, \omega^2\}$ under multiplication, where $\omega = e^{2\pi i/3}$.

Solution

The homomorphism $f : \mathbb{Z} \to \mathbb{C}^\times$, $n \mapsto \omega^n$, has image $\{1, \omega, \omega^2\}$ and kernel $3\mathbb{Z}$, because $\omega^n = 1$ exactly when $3 \mid n$. Therefore $\mathbb{Z}/3\mathbb{Z} \cong \operatorname{im} f$.

18.3 Commutators and the derived subgroup

Definition 18.4. The *commutator* of $x, y \in G$ is $[x, y] = xyx^{-1}y^{-1}$. The *derived subgroup* $D(G) = [G, G]$ is the subgroup generated by all commutators.

The equality $[x, y] = e$ is equivalent to $xy = yx$. Thus G is abelian iff $D(G) = \{e\}$. Every group homomorphism $f : G \to H$ sends $[x, y]$ to $[f(x), f(y)]$, so $f(D(G)) \subseteq D(H)$. In particular every automorphism preserves $D(G)$, making it *characteristic*; characteristic subgroups are normal because inner conjugations are automorphisms. Directly, $g[x, y]g^{-1} = [gxg^{-1}, gyg^{-1}]$ also proves normality.

Practice 18.5. Prove that $G/D(G)$ is abelian.

Solution

For $x, y \in G$, the commutator of their cosets is $[xD(G), yD(G)] = [x, y]D(G) = D(G)$, the identity. Hence all cosets commute.

18.4 Abelianization: universal property, functor, adjunction

Definition 18.6. The *abelianization* of G is $G_{\mathrm{ab}} = G/D(G)$, with quotient map $\eta_G : G \to G_{\mathrm{ab}}$.

If A is abelian and $f : G \to A$ a group morphism, then $f([x, y]) = e$ for every x, y, so $D(G) \subseteq \ker f$. Quotient factorization therefore gives a unique morphism $\bar{f} : G_{\mathrm{ab}} \to A$ with $f = \bar{f}\eta_G$. This is the universal property: abelianization is the most general abelian quotient of G. For a homomorphism $f : G \to H$, the inclusion $f(D(G)) \subseteq D(H)$ gives a well-defined induced map

$$f_{\mathrm{ab}} : G_{\mathrm{ab}} \to H_{\mathrm{ab}}, \qquad gD(G) \mapsto f(g)D(H).$$

Identity and composition are preserved because their induced maps agree on every coset: $(\mathrm{id}_G)_{\mathrm{ab}} = \mathrm{id}_{G_{\mathrm{ab}}}$ and $(hf)_{\mathrm{ab}} = h_{\mathrm{ab}} f_{\mathrm{ab}}$. Hence abelianization is a functor $\mathbf{Grp} \to \mathbf{Ab}$. The quotient maps are natural: $f_{\mathrm{ab}}\eta_G = \eta_H f$. Let $I : \mathbf{Ab} \hookrightarrow \mathbf{Grp}$ be the inclusion. The universal property is a natural bijection

$$\mathrm{Hom}_{\mathbf{Ab}}(G_{\mathrm{ab}}, A) \;\cong\; \mathrm{Hom}_{\mathbf{Grp}}(G, I(A)), \qquad \bar{f} \longmapsto \bar{f}\eta_G.$$

Thus abelianization is left adjoint to inclusion. Naturality in G is precomposition with a group map; naturality in A is postcomposition with an abelian-group map. Both commute with quotient factorization because induced maps are unique.

Practice 18.7. Compute the abelianization of $F(x, y)$ and of an abelian group A.

Solution

The free group $F(x, y)$ becomes free abelian on x, y after killing commutators, so its abelianization is $\mathbb{Z}^2$, recording the two exponent sums. For abelian A, every commutator is e, so $D(A) = \{e\}$ and $A_{\mathrm{ab}} = A/\{e\} \cong A$.

18.5 Abelianization and rank

Let $[G, G]$ be generated by commutators $ghg^{-1}h^{-1}$. The abelianization $G_{\mathrm{ab}} = G/[G, G]$ is preserved by isomorphism, and

$$F(A)_{\mathrm{ab}} \cong \bigoplus_{a \in A} \mathbb{Z}.$$

If $F(A) \cong F(B)$ and $|A| = n < \infty$, abelianizing and tensoring with $\mathbb{Q}$ gives $\mathbb{Q}^n \cong \bigoplus_B \mathbb{Q}$. Vector-space dimension implies B is finite and $|B| = n$. This solves Problem 12.

Checkpoint: Lectures 12–13

Prove the equivalent normality conditions; check quotient multiplication using representatives; factor a map through a quotient and determine when the induced map is injective/surjective; use $G/\ker f \cong \operatorname{im} f$; prove commutators vanish in every abelian target; compute induced abelianization maps.

Chapter 19

Proof checklist and mock exam

Claim	Obligation
Injective	equal outputs imply equal inputs
Surjective	construct a preimage for an arbitrary target
Equivalence	reflexive, symmetric, transitive
Representative formula	well-definedness first
Group	closure, associativity, identity, every inverse
Subgroup	nonempty and ab^{-1} closed
Homomorphism	compare $f(ab)$ with $f(a)f(b)$
Isomorphism	homomorphism, injective, surjective
Universal object	construct forced maps and prove uniqueness
Presented group free	mutually inverse maps or valid Tietze moves

19.1 Skills checkpoints

Checkpoint: Sets and mappings

Without notes, you should be able to count injections in every size case; prove the 2-adic bijection; characterize when all permutations commute; check all three equivalence axioms; describe fraction, difference, rational-translation, and residue classes; prove the composite-map result; and recognize complement as an involution.

Checkpoint: Operations and groups

For every proposed structure: verify closure; expand both sides for associativity; swap variables for commutativity; solve for an identity; solve for inverses. For finite laws count free table cells. For function spaces check each identity pointwise.

Problem	Key answer
1(a)	neither associative nor commutative
1(b)	neither
1(c)	commutative, generally not associative
1(d)	associative, generally not commutative
2	the displayed two-element table is a counterexample
3	n^{n^2} and $n^{n(n+1)/2}$
4	adjoin a formal identity
5	neither is a group
6	use $z^{-1} = z^{n-1}$
7	check pointwise

Checkpoint: Quotients and morphisms

Problems 1–3: prove well-definedness, then axioms, then uniqueness from the projection. Problems 4–6: use the one-step subgroup test and keep the ambient operation visible. Problems 7–11: test identity first, prove preservation, and use quotient/product universal properties. Problems 12–14: abelianization detects rank; Schreier plus Nielsen handles the rank-three subgroup; solve for and eliminate a generator in a presentation.

No.	Proof spine
1	Division algorithm; $[a] + [b] = [a + b]$; identity and inverse; projection forces uniqueness.
2	$f_{a,b} f_{c,d} = f_{ac,ad+b}$ and $f_{a,b}^{-1} = f_{1/a,-b/a}$.
3	Repeat Problem 1 with differences in $\mathbb{Z}$.
4	No, yes, yes, yes.
5	Powers versus multiples.
6	Stabilizer is a subgroup; the transporter misses identity.
7	Yes, no, yes, no, yes.
8	Put $a = f(1)$.
9	Expand the cube identity and cancel.
10	Factor coordinate projections, pair them, prove bijective.
11	Exponential; divisibility obstruction for rationals.
12	Abelianize and compare rational dimensions.
13	Index-two kernel, Schreier basis, Nielsen move.
14	Eliminate x.

19.2 Cumulative mock exam

1. Let $|A| = 3, |B| = 5$. Count maps and injections $A \to B$.
2. Prove the fraction relation on $\mathbb{Z} \times \mathbb{Z}_{\neq 0}$ is equivalence and identify its quotient with $\mathbb{Q}$.
3. On $\mathbb{R}$, set $a * b = a + b - 2ab$. Test associativity and find an identity.
4. Prove $\{f \in \mathrm{Sym}(X) : f(A) = A\}$ is a subgroup.
5. Show $\phi : \mathbb{Z}/12\mathbb{Z} \to \mathbb{Z}/4\mathbb{Z}$, $[a]_{12} \mapsto [a]_4$, is a surjective homomorphism; find its kernel.
6. Prove evaluation at x_0 from real-valued functions on X to $(\mathbb{R}, +)$ is surjective.
7. Prove $(G \times H)/(N \times M) \cong (G/N) \times (H/M)$ whenever the quotients are defined.

8. Simplify $\langle a, b, c \mid c^{-1}ab^2 = e \rangle$.

Answers

1. $5^3 = 125$ and $5 \cdot 4 \cdot 3 = 60$.
2. Use invariant $(a, b) \mapsto a/b$; the induced map on classes is well-defined and bijective.
3. Both parenthesizations equal $a + b + c - 2ab - 2ac - 2bc + 4abc$; identity is 0.
4. Identity preserves A and f, g preserving A implies fg^{-1} does too.
5. Divisibility by 12 implies divisibility by 4; kernel $\{[0], [4], [8]\}$.
6. The constant function with value r maps to arbitrary r.
7. $[(g, h)] \mapsto (gN, hM)$ is a well-defined bijective homomorphism.
8. $c = ab^2$, so eliminate c and obtain $F(a, b)$.

Review strategy

First reproduce every definition from memory. Redo Assignment 2's operation checks and Assignment 3 Problems 1, 7, and 10; they expose the common gaps of associativity, well-definedness, and universal properties. Finally rehearse abelianization for Problem 12, Schreier–Nielsen for 13, and generator elimination for 14.